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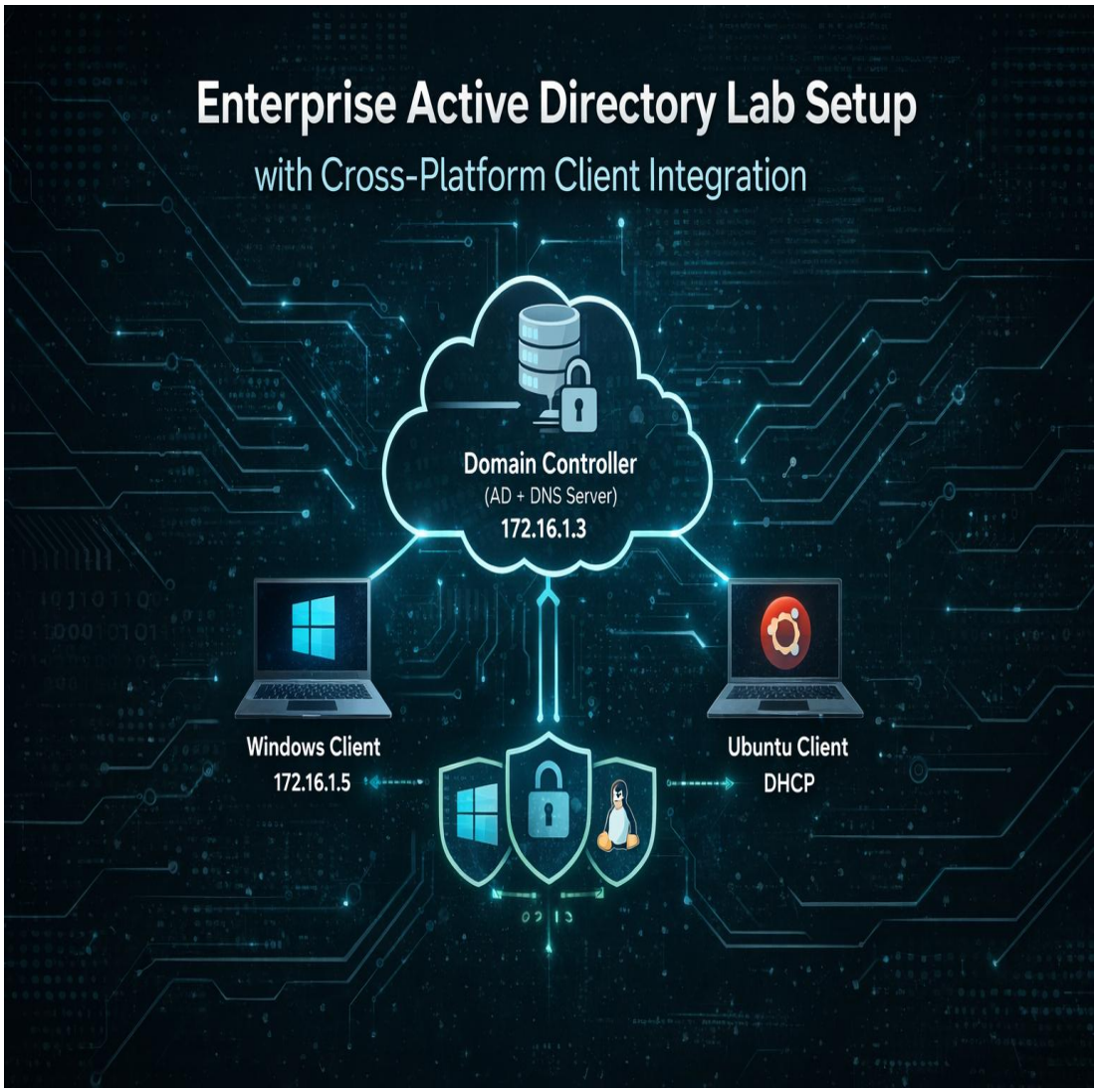

Active Directory Lab Setup with Cross-Platform Client Integration (Windows & Linux)

RITIK PATEL

SOC – Threat Research Intern

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1. Introduction

Active Directory (AD) is a centralized identity and access management system widely used in enterprise environments to manage users, computers, and resources within a network. It allows organizations to enforce security policies, control authentication, and streamline administrative tasks from a single point.

In modern IT infrastructures, Active Directory plays a critical role in ensuring secure access to systems and services. By using domain-based authentication, organizations can manage multiple devices and users efficiently while maintaining strong security controls.

In this project, a complete Active Directory lab environment was built using a Windows Server as a Domain Controller. Multiple client machines, including a Windows 10 system and an Ubuntu machine, were integrated into the domain. This setup demonstrates how centralized authentication works across different operating systems.

The objective of this lab was to:

- Configure a Domain Controller with Active Directory and DNS
- Create and manage domain users
- Join Windows and Linux systems to the domain
- Validate authentication across different clients

This hands-on implementation provides a foundational understanding of how enterprise environments manage identity, access, and system integration.

2. Lab Architecture

To simulate a real-world enterprise environment, a virtual lab was designed using multiple machines interconnected within the same network. The setup consists of a Domain Controller and multiple client systems integrated into the domain.

Environment Overview

The lab environment includes the following components:

1. Windows Server (Domain Controller)
    - Role: Active Directory Domain Services (AD DS) and DNS Server
    - IP Address: 172.16.1.3
    - Domain Name: RIT.local
2. Windows 10 Client
    - Role: Domain-joined endpoint
    - Used to simulate user authentication and activity
    - IP Address: 172.16.1.5
3. Ubuntu Client
    - Role: Linux system integrated with Active Directory
    - Used to demonstrate cross-platform authentication
    - IP Address: Assigned via DHCP

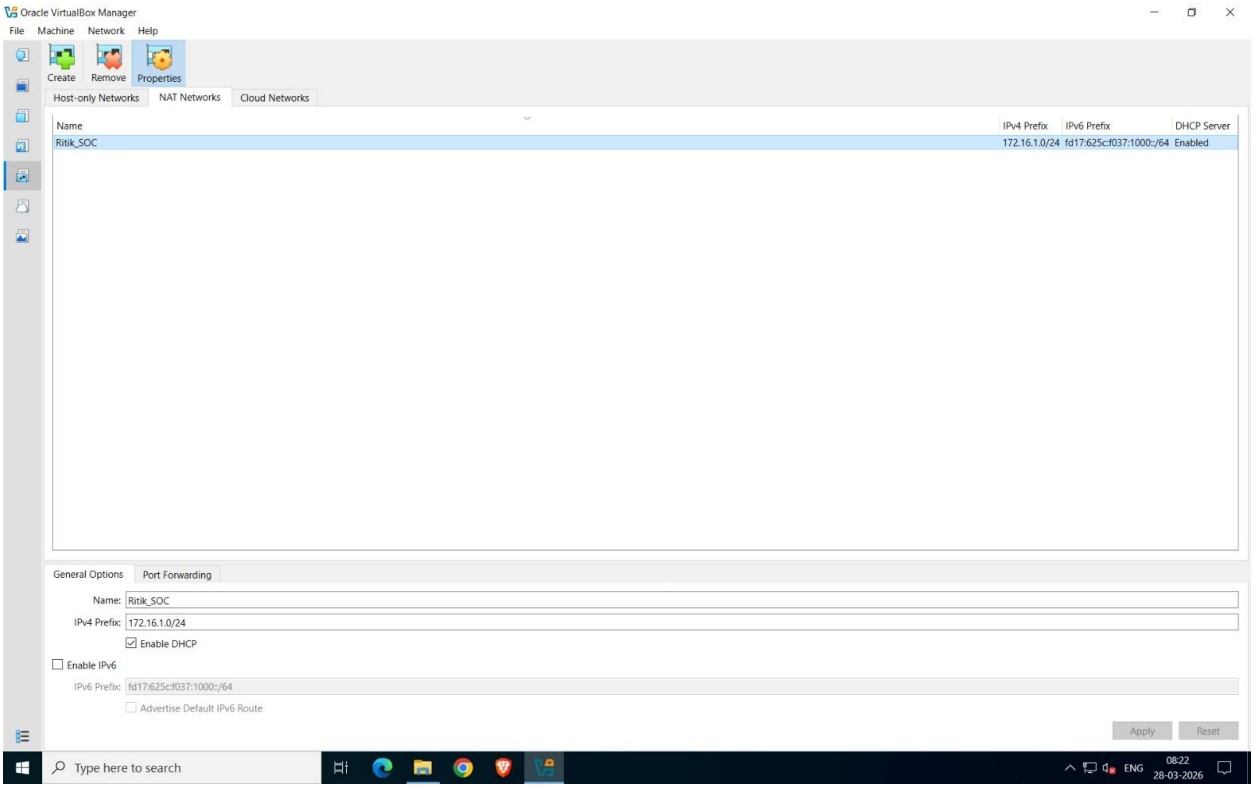
Network Configuration

All machines are connected within the same virtual network using **VirtualBox NAT Network**, enabling communication between systems.

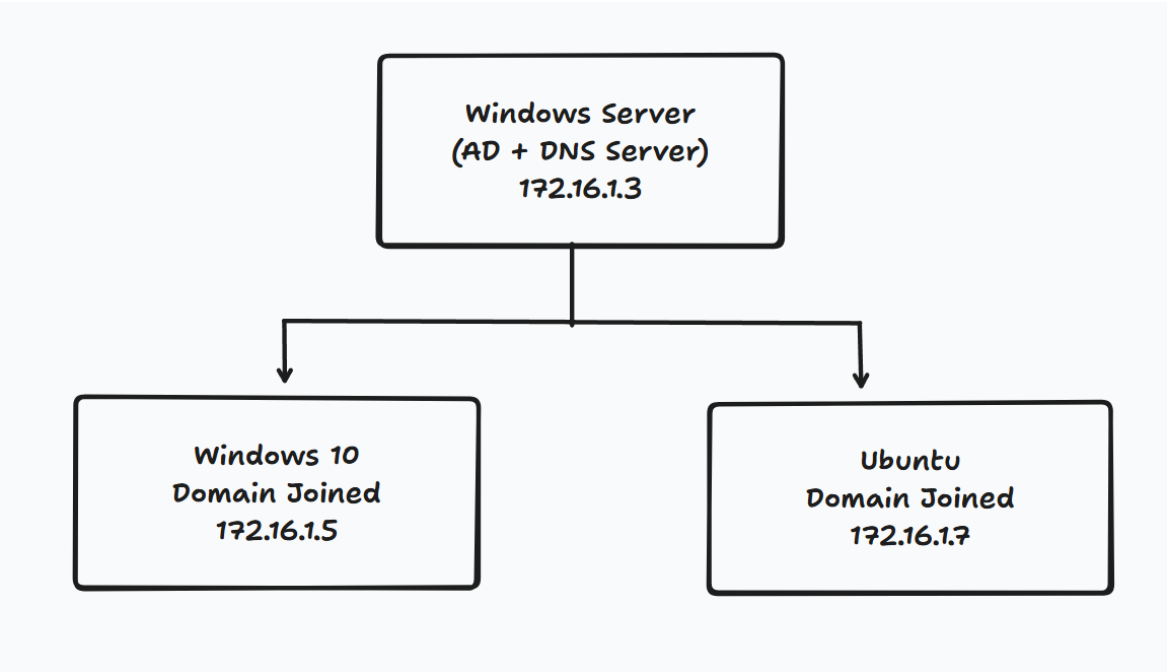
- Network Range: 172.16.1.0/24
- Gateway: 172.16.1.1

- DNS Server: 172.16.1.3 (Domain Controller)

The Domain Controller also acts as the DNS server, which is critical for domain name resolution and successful domain joining.



Logical Architecture



Key Objectives of the Architecture

- Enable centralized authentication using Active Directory
- Integrate both Windows and Linux systems into the domain
- Simulate real enterprise network behavior
- Demonstrate cross-platform identity management

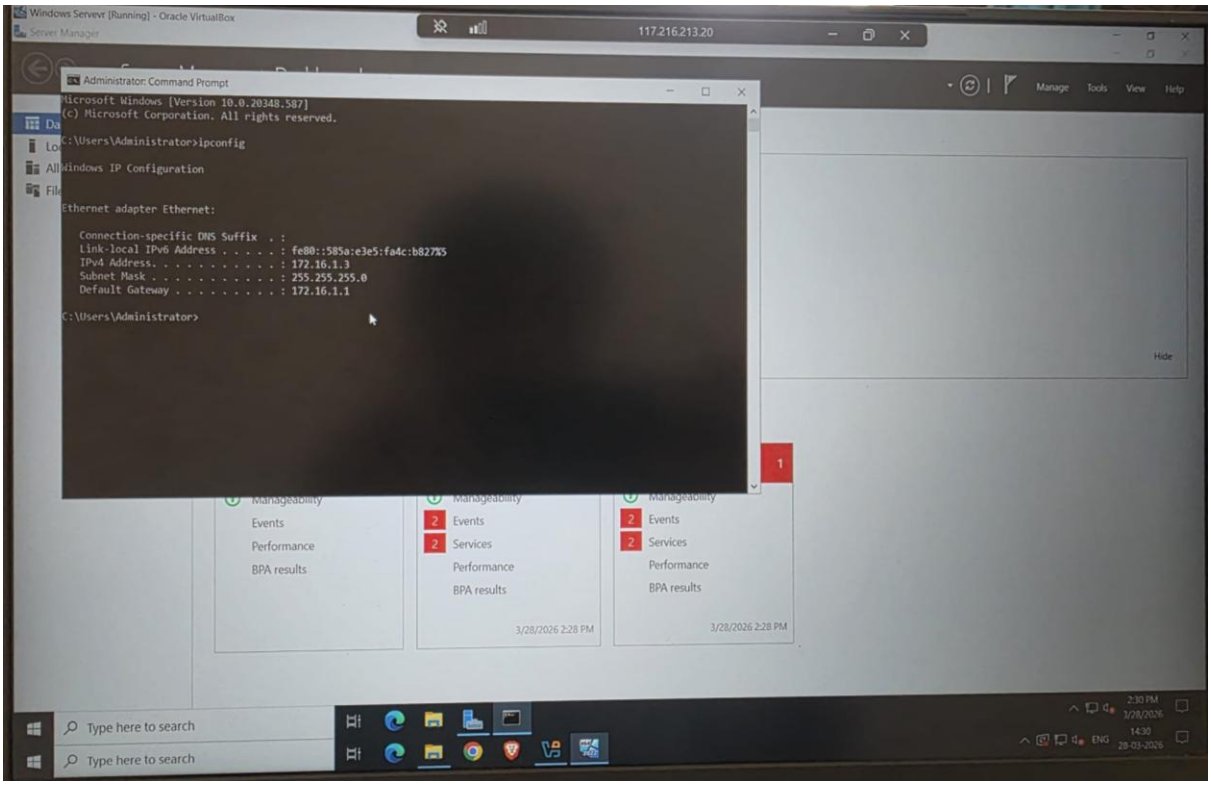
3. Active Directory Setup

The first step in building the lab environment was configuring a Windows Server as a Domain Controller by installing and setting up Active Directory Domain Services (AD DS).

Windows Server Installation

A Windows Server virtual machine was created using VirtualBox and configured with appropriate hardware resources. After installation, the system was prepared for domain configuration.

- Verified network connectivity
- Assigned a static IP address (172.16.1.3)
- Configured the system hostname



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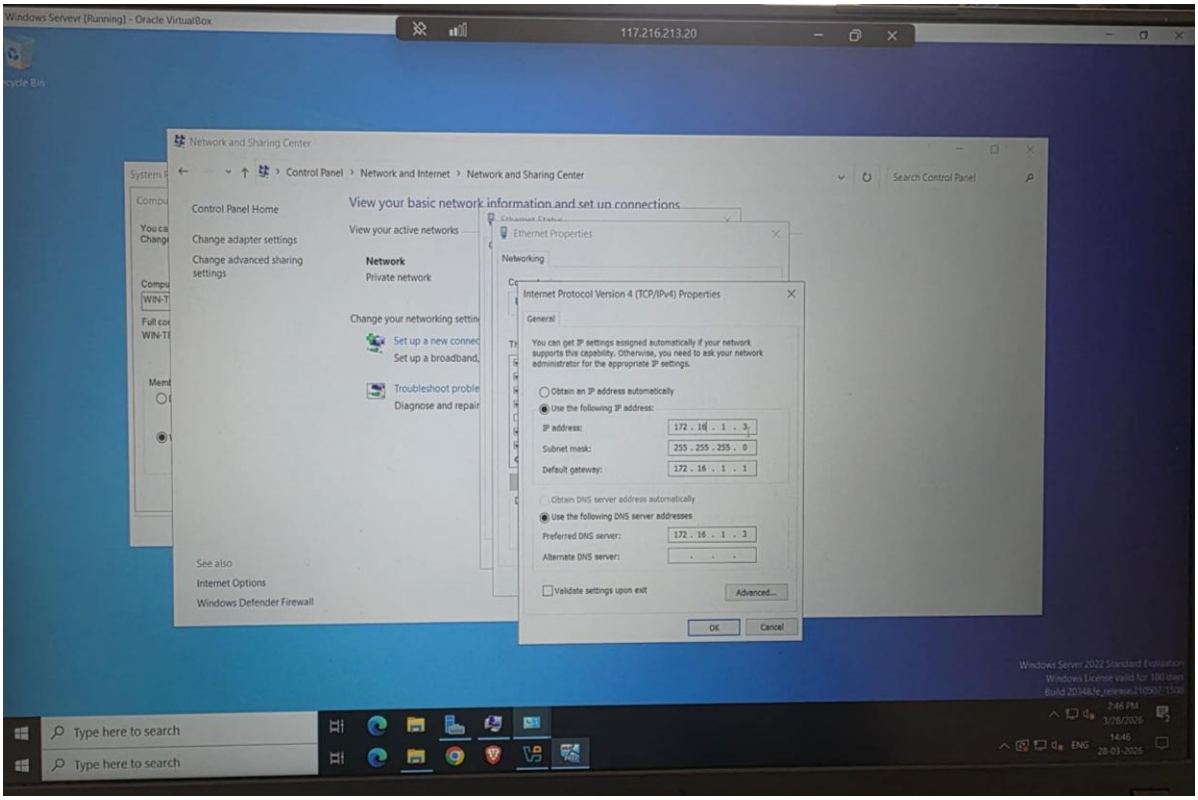

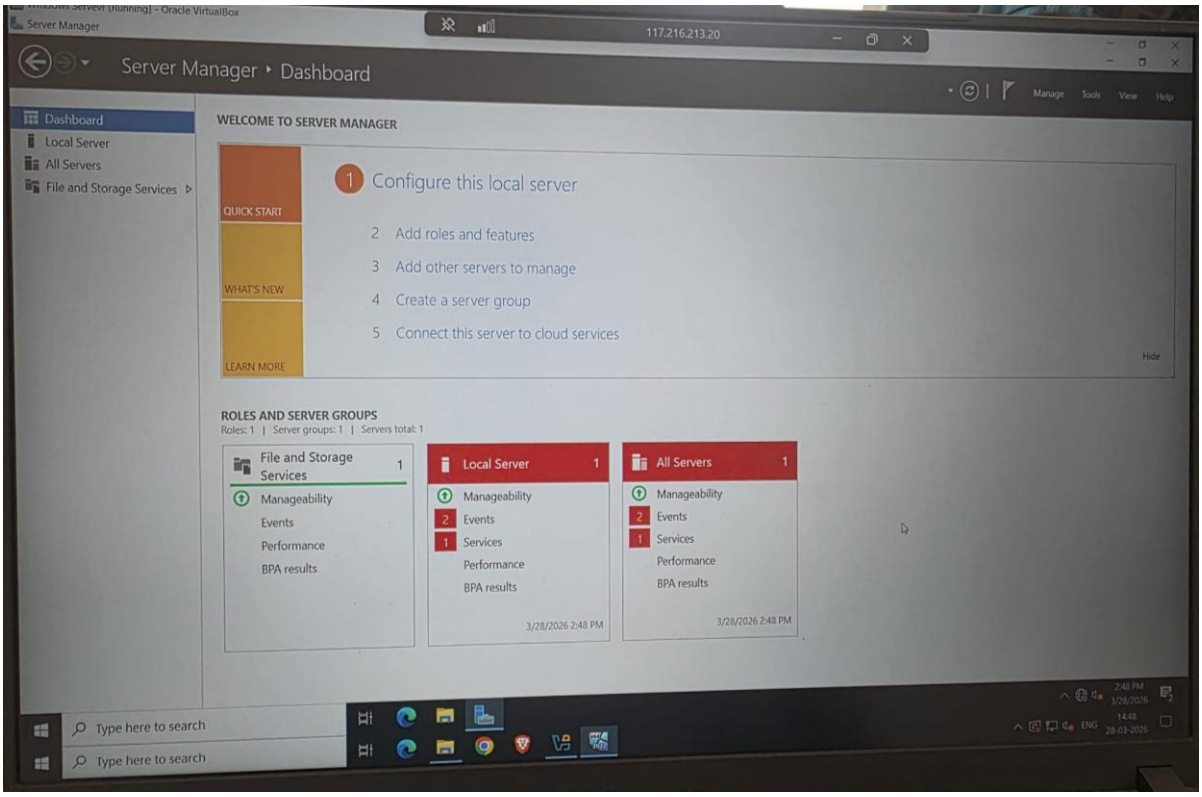

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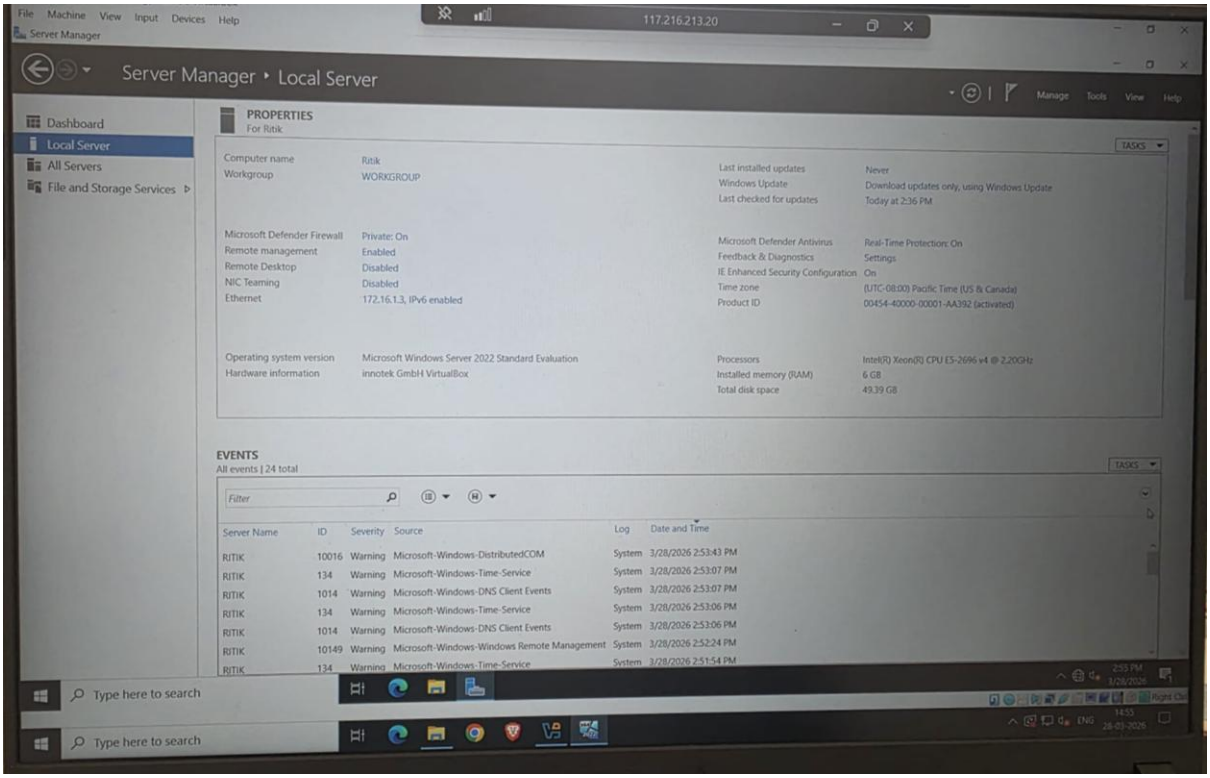

Installing Active Directory Domain Server (AD DS)

Active Directory Domain Services was installed using the Server Manager.

Steps performed:

1. Opened **Server Manager**
2. Selected **Add Roles and Features**
3. Completed the installation process

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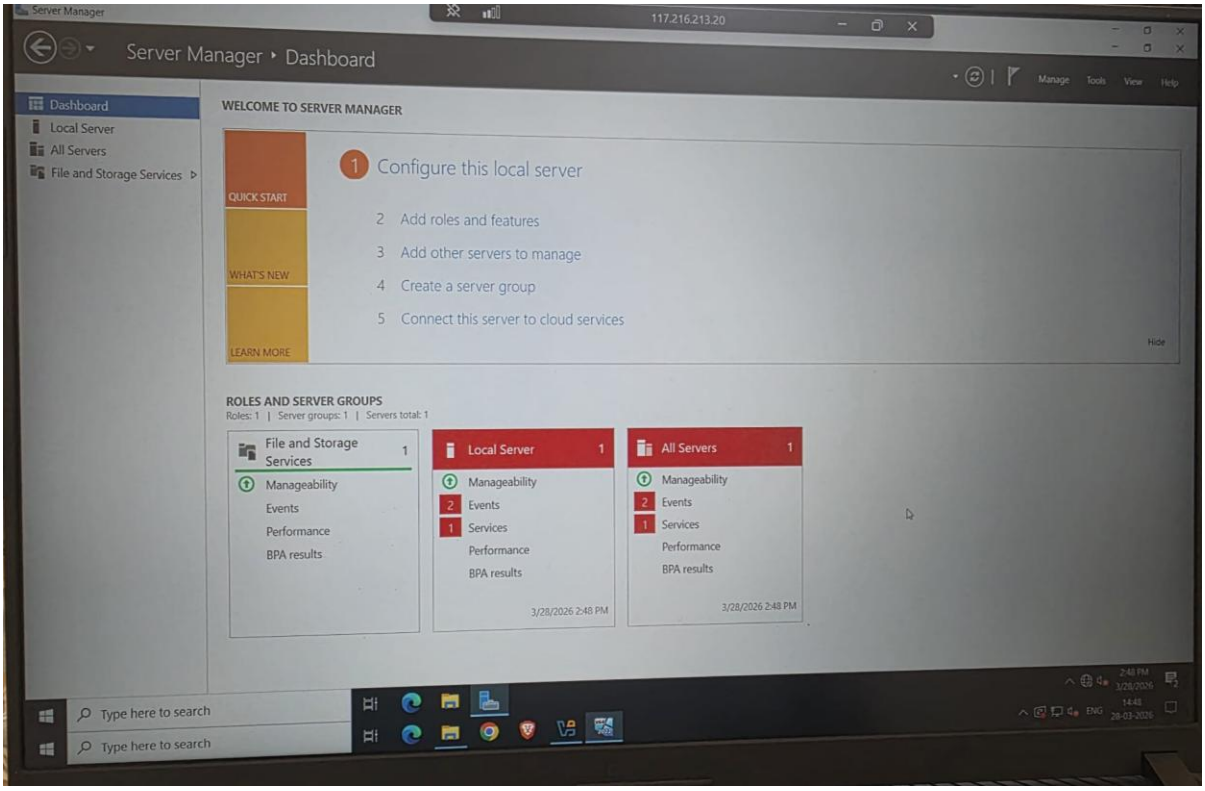

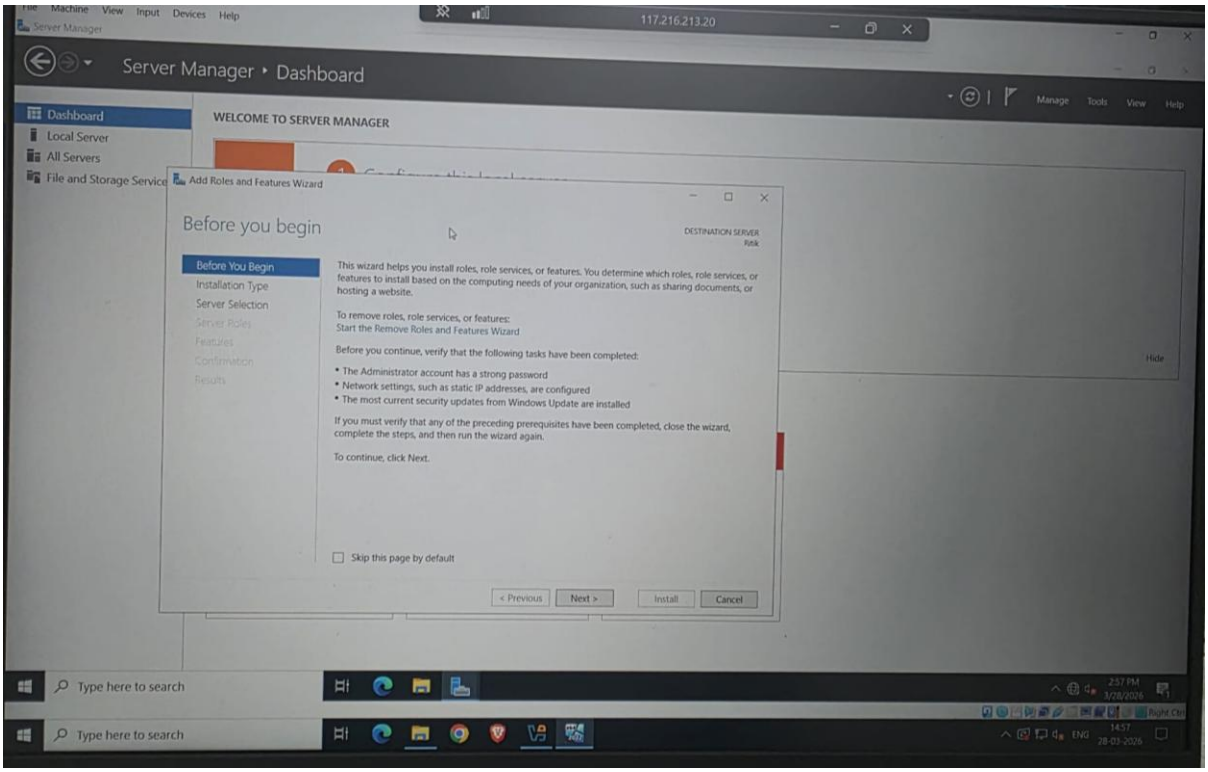

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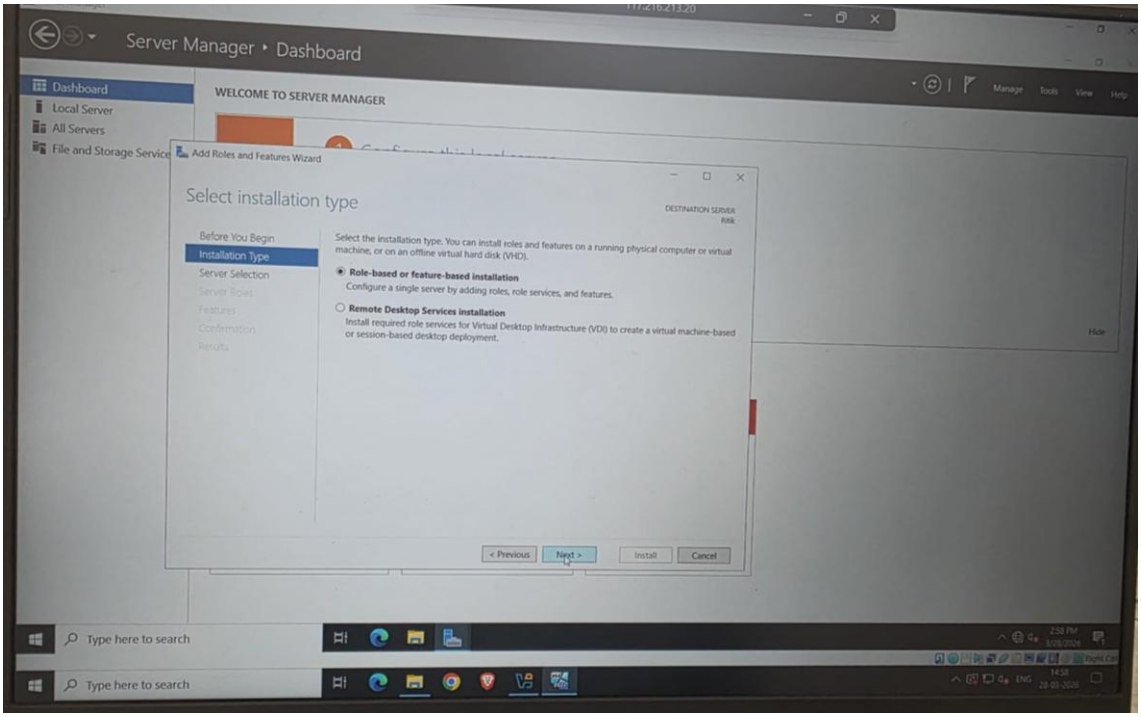

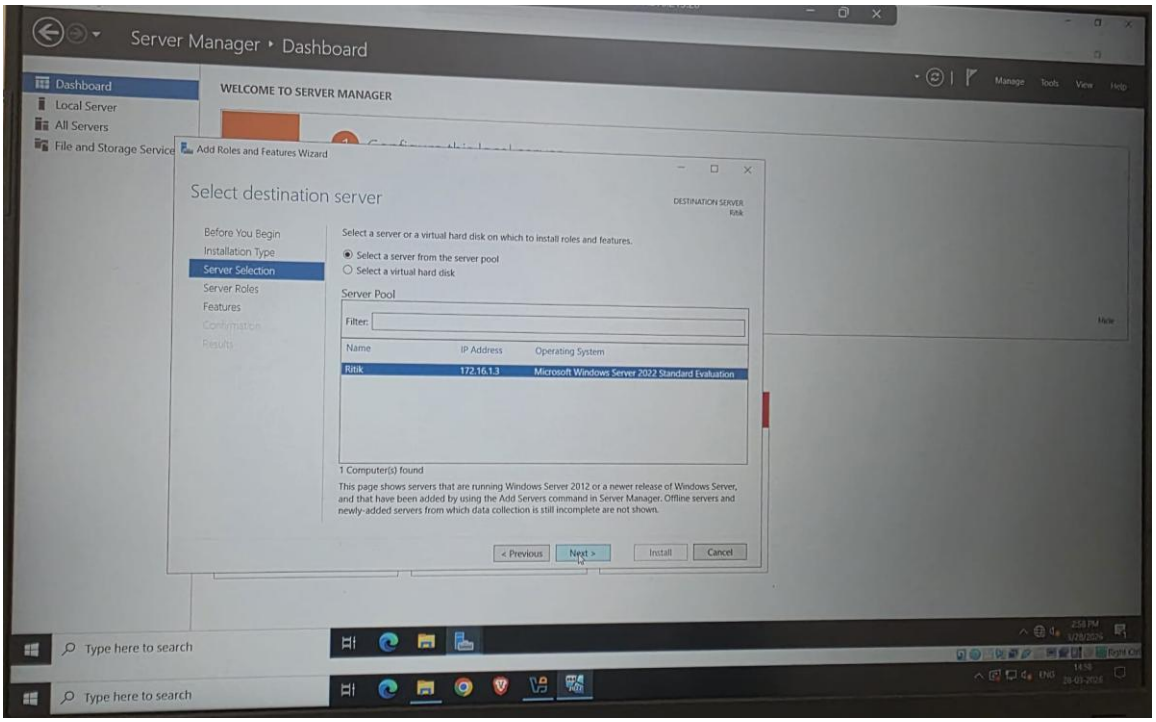

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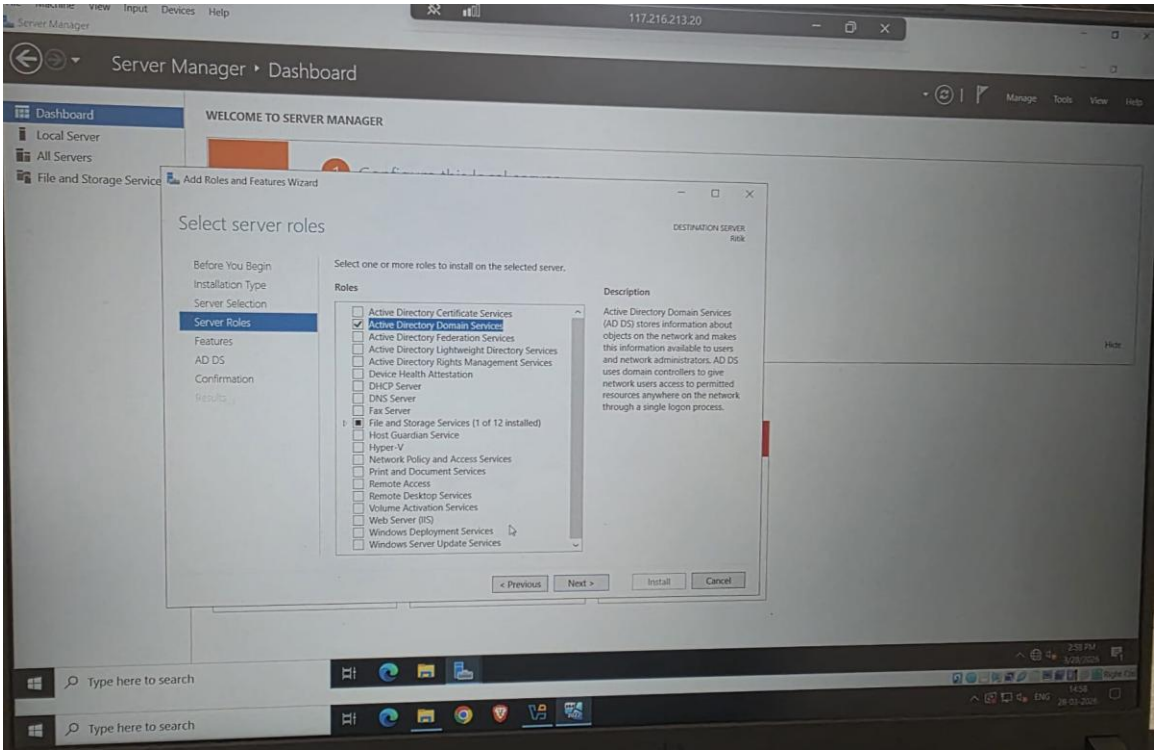

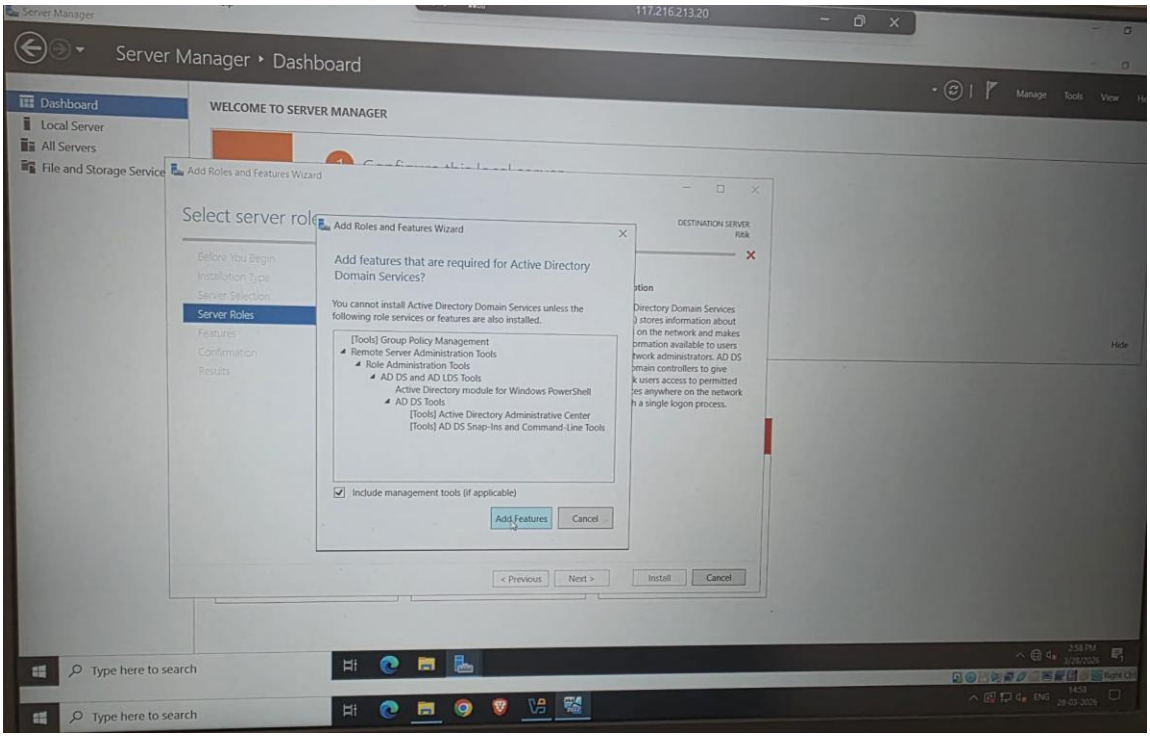

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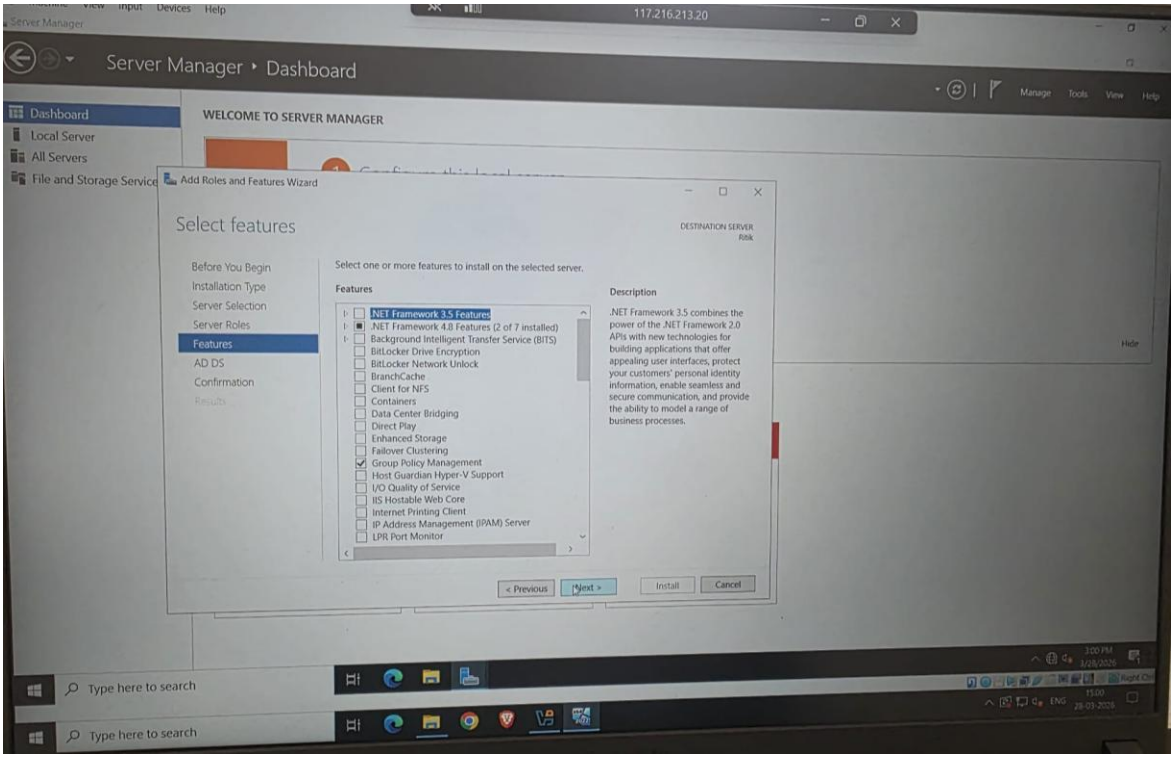

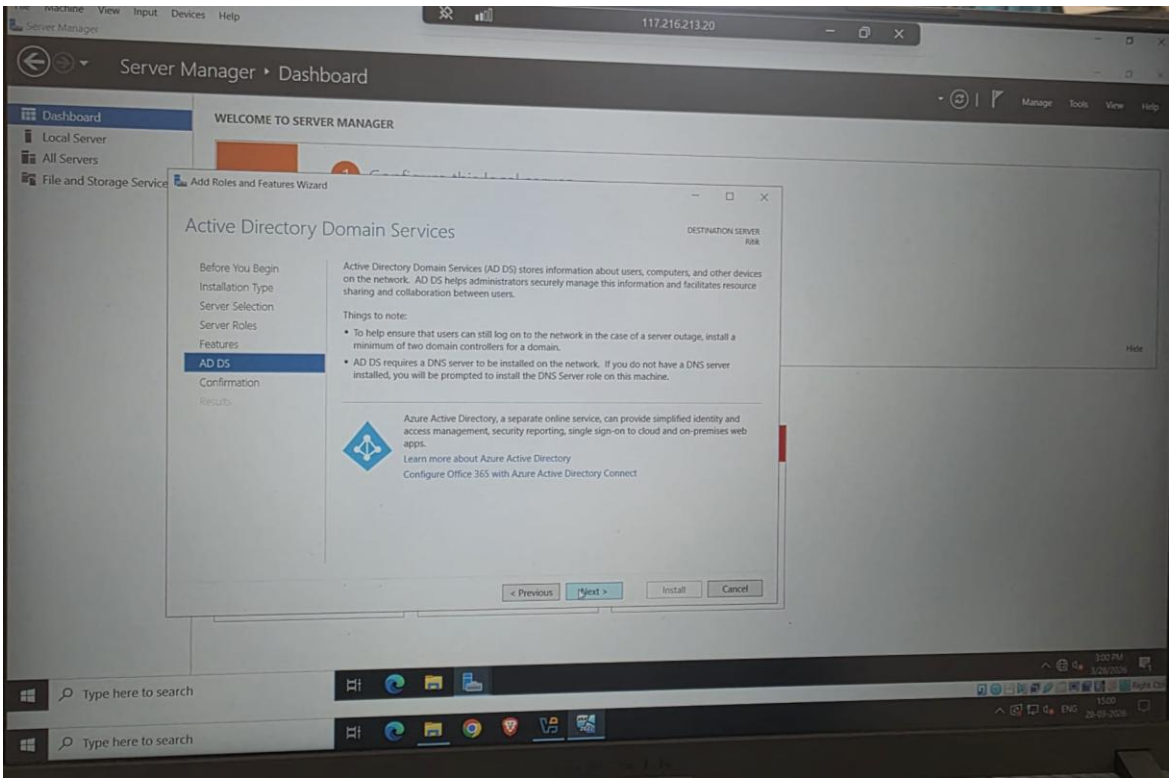

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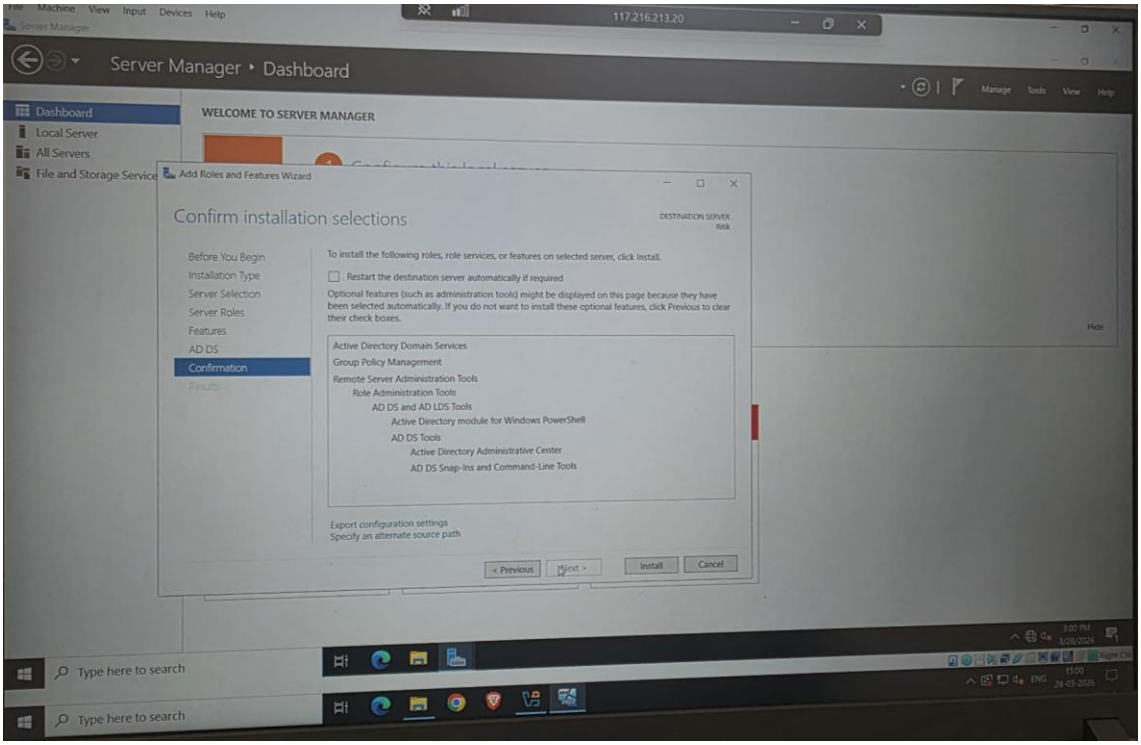

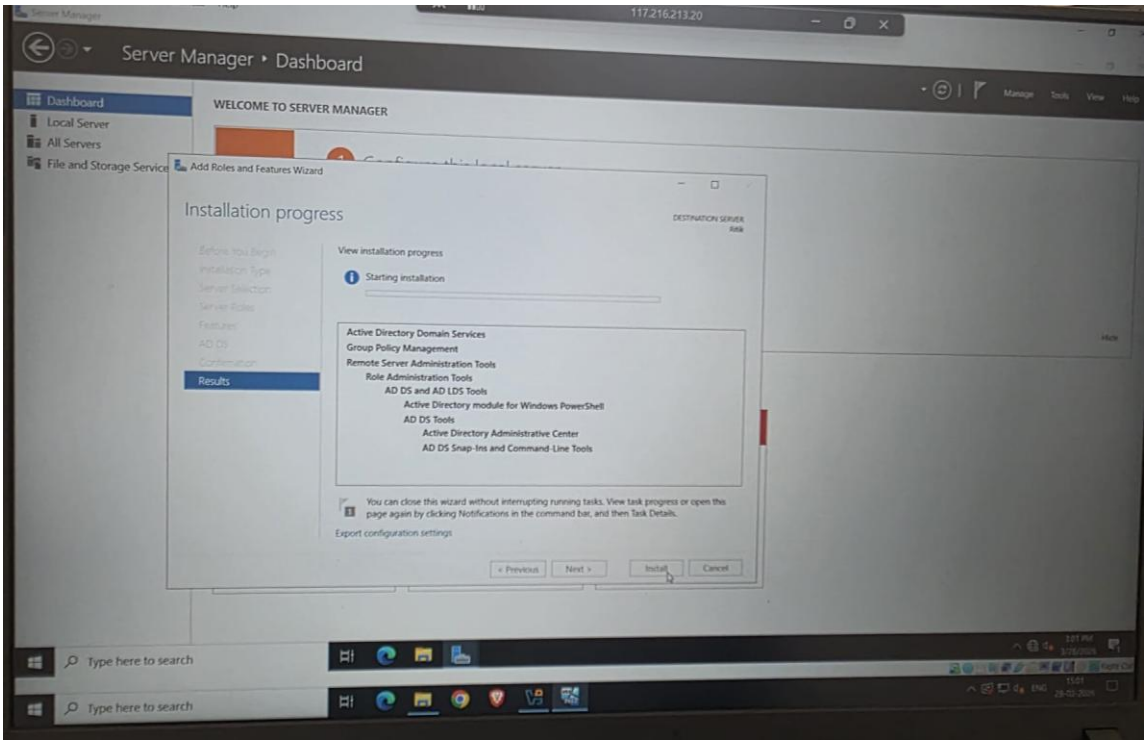

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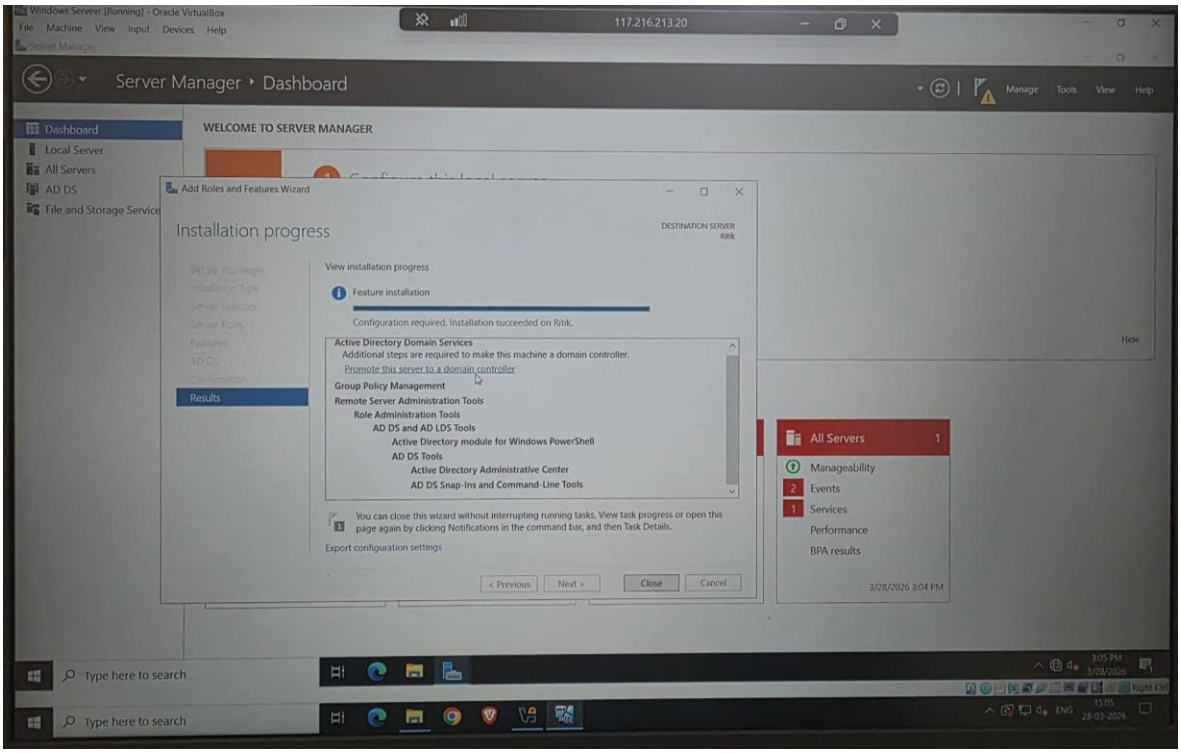

Once you have reached here click on “Promoting this server to a domain controller”

Promoting Server to Domain Controller

After installing AD DS, the server was promoted to a Domain Controller.

Key configuration:

- Created a **new forest**
- Domain name: RIT.local
- Enabled DNS server during setup
- Configured Directory Services Restore Mode (DSRM) password

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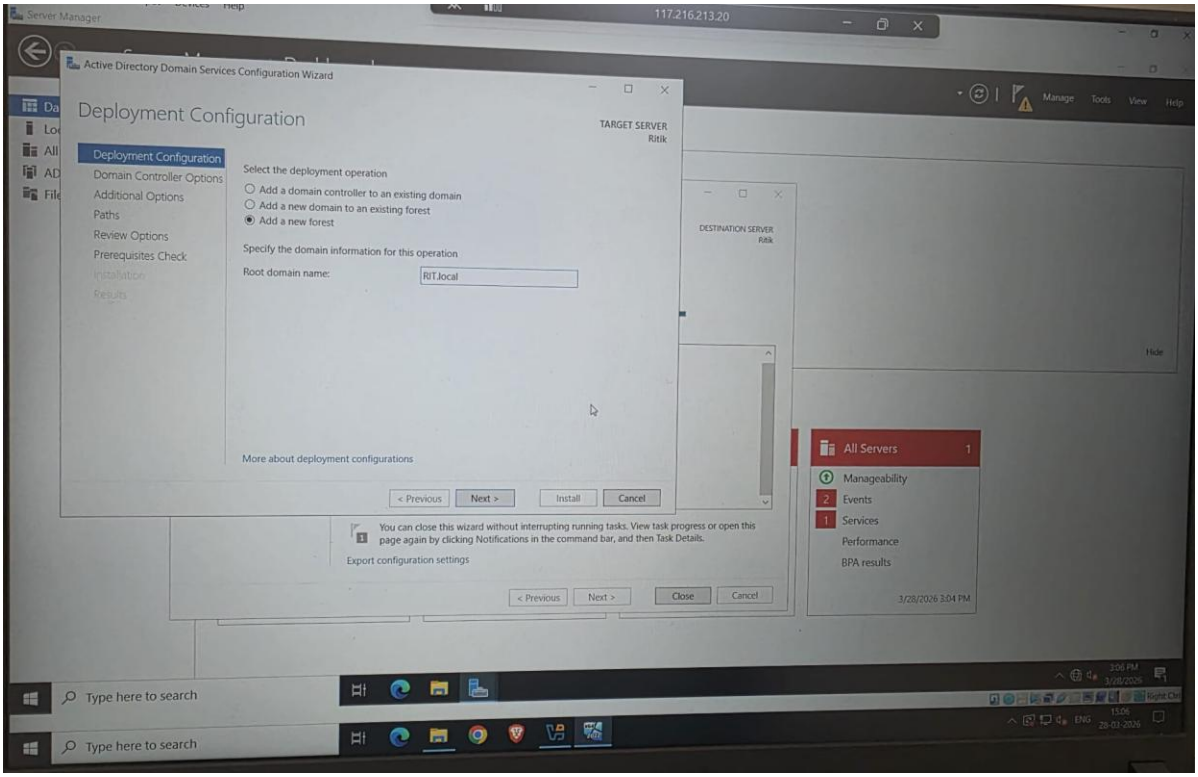

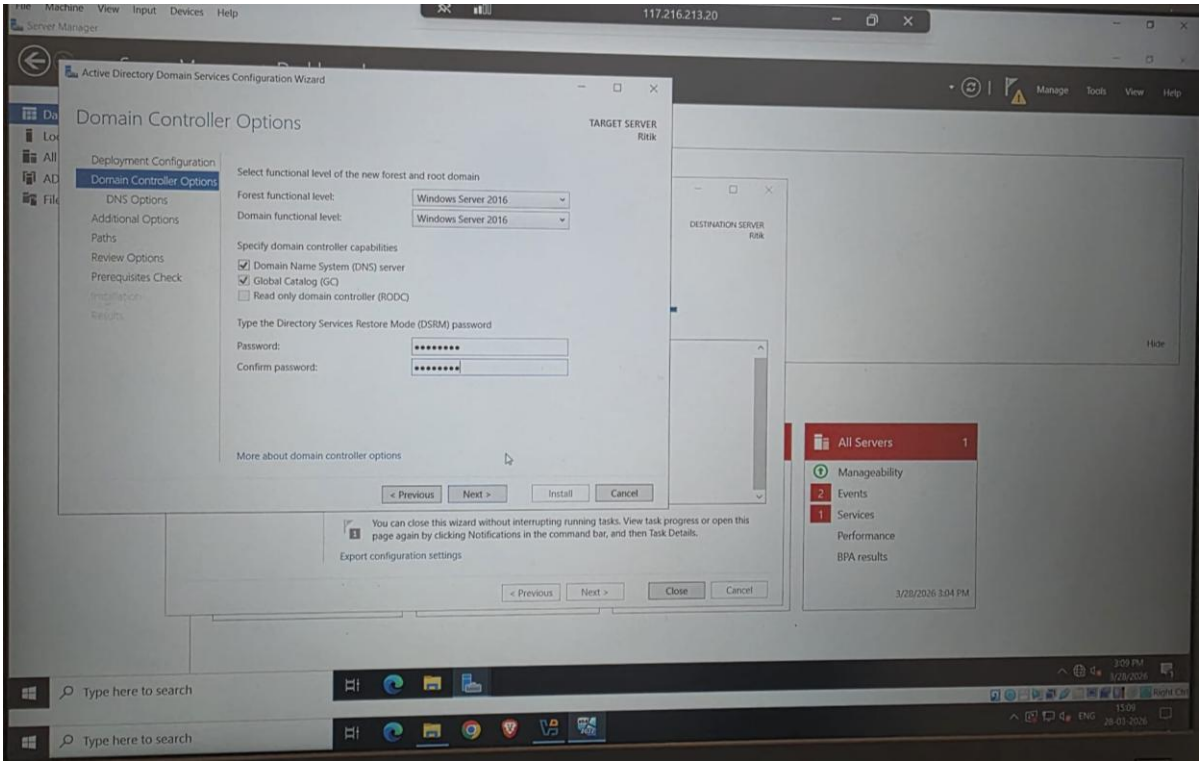

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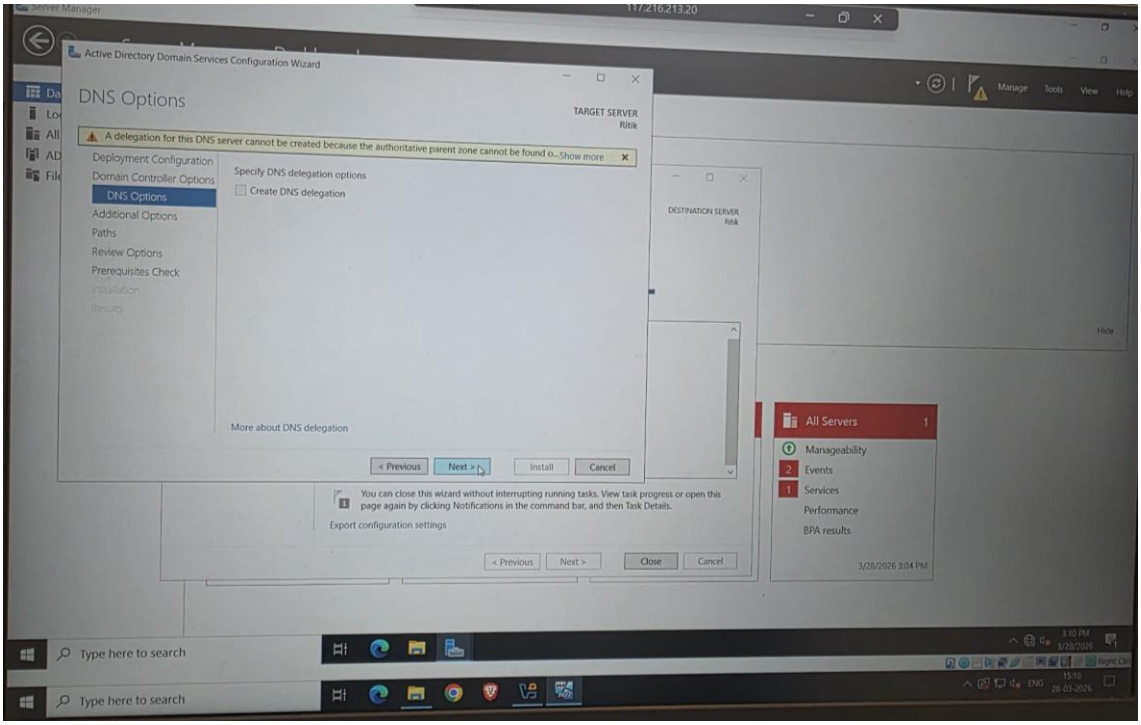

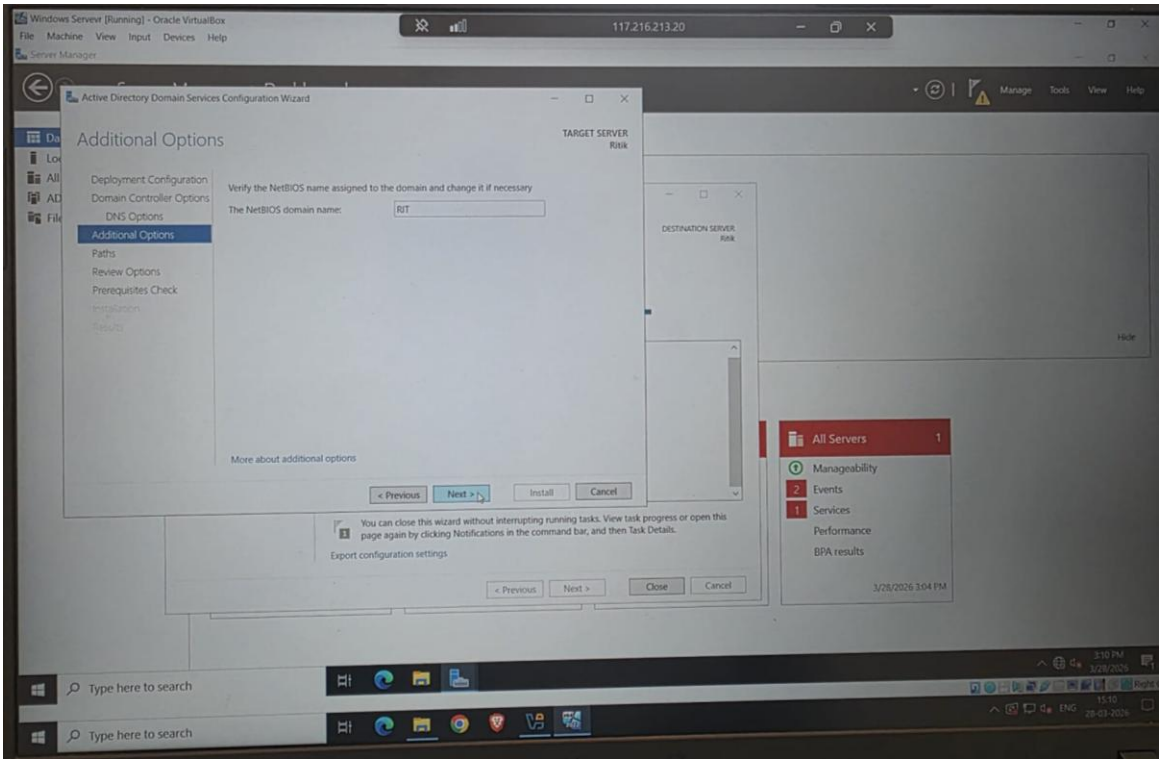

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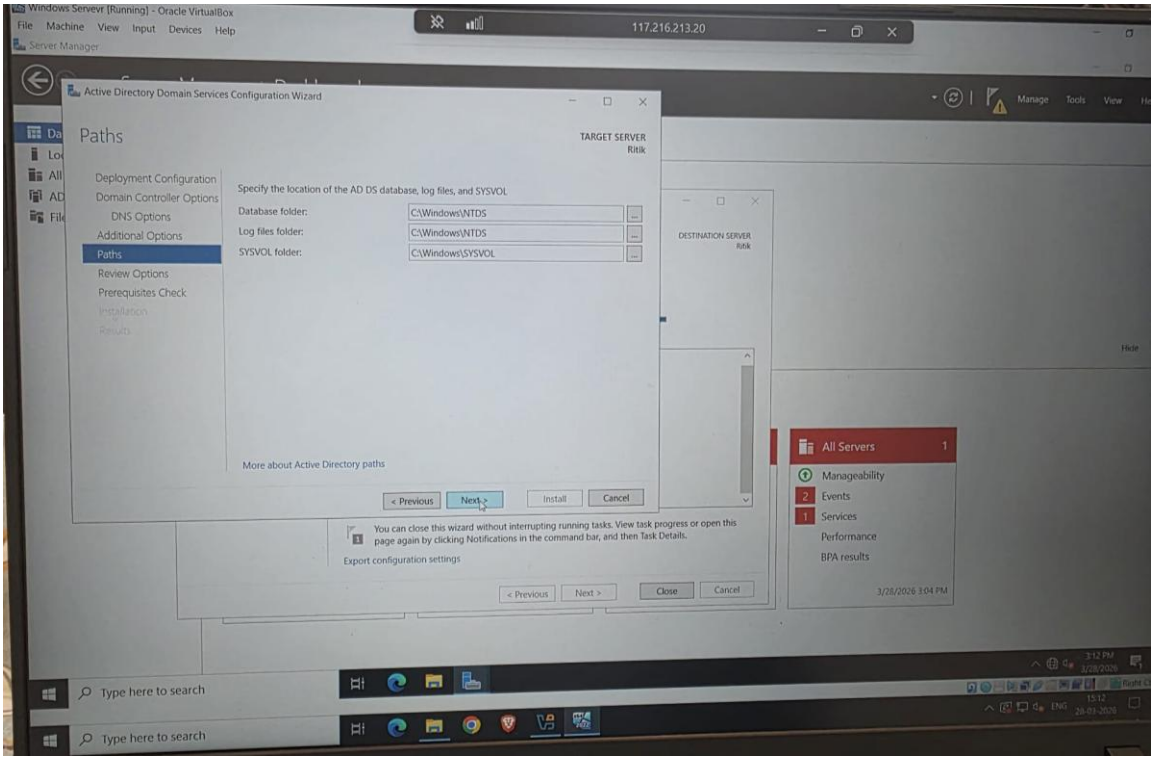

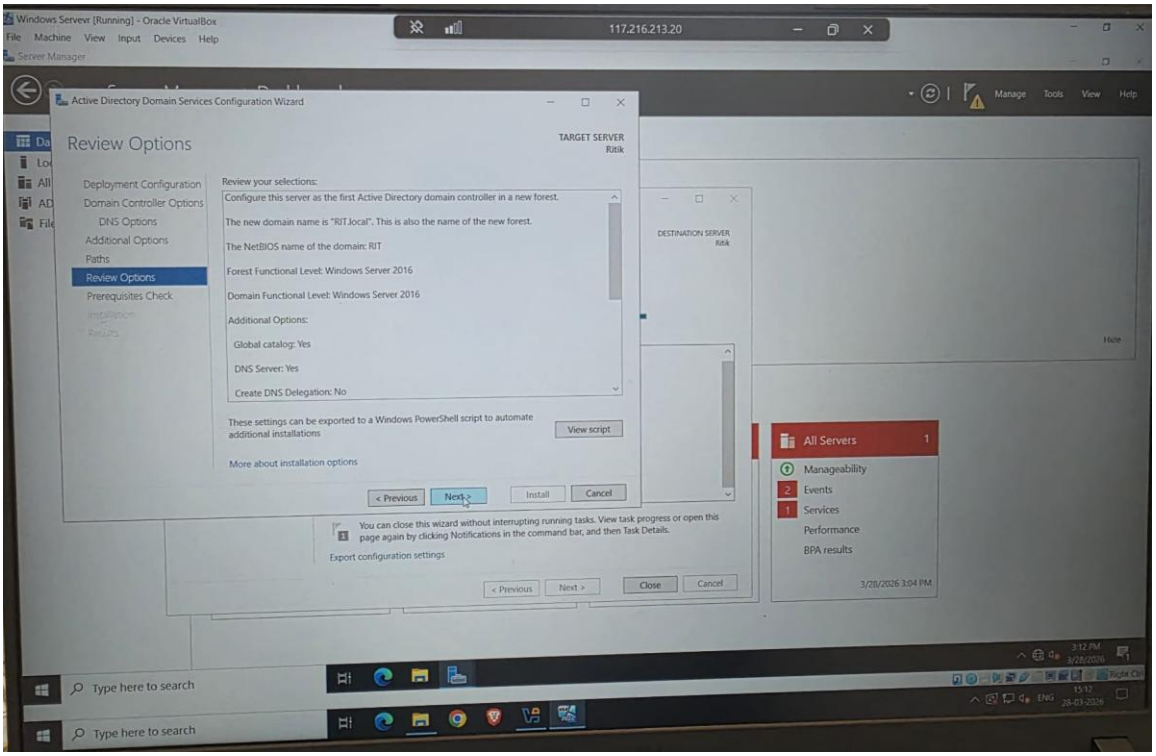

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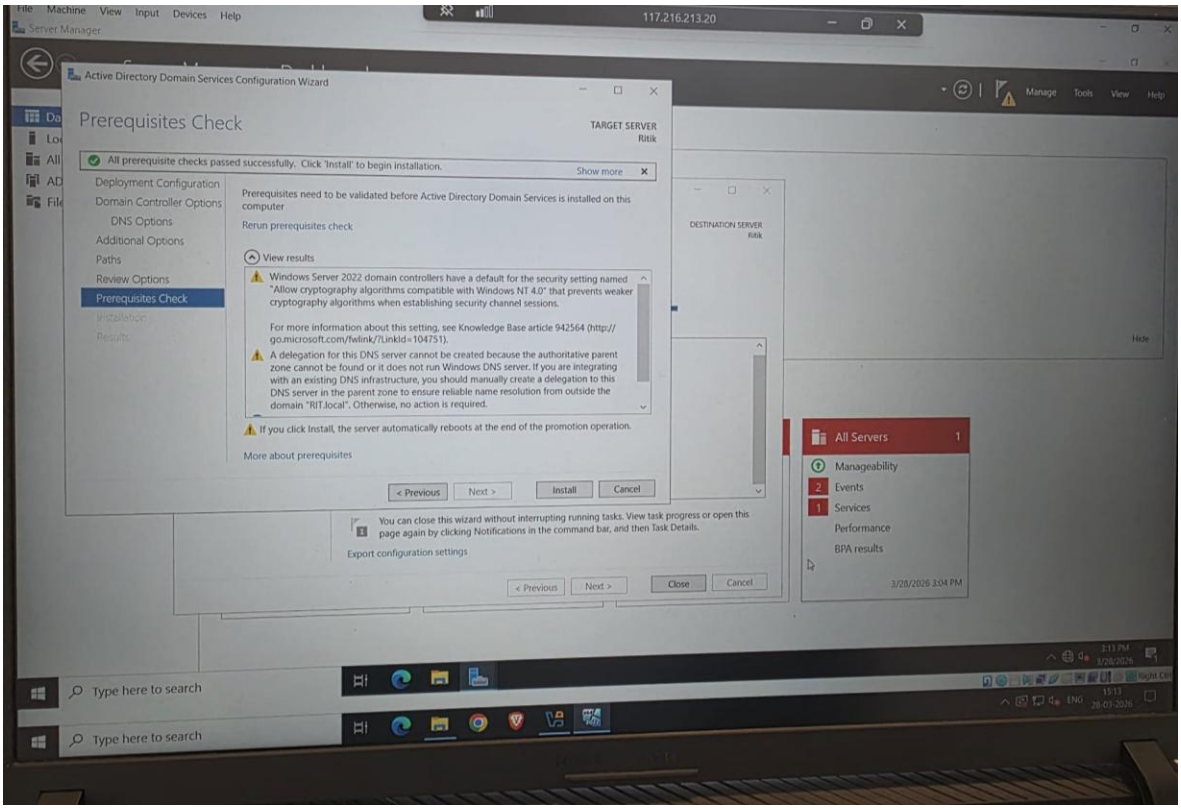


Once the configuration was complete, the server automatically restarted and became the primary Domain Controller for the environment.

DNS Configuration

During the domain controller setup, the DNS role was automatically installed and configured.

- DNS service used for domain name resolution
- All domain clients rely on this server (172.16.1.3) for authentication
- Ensures proper communication between systems

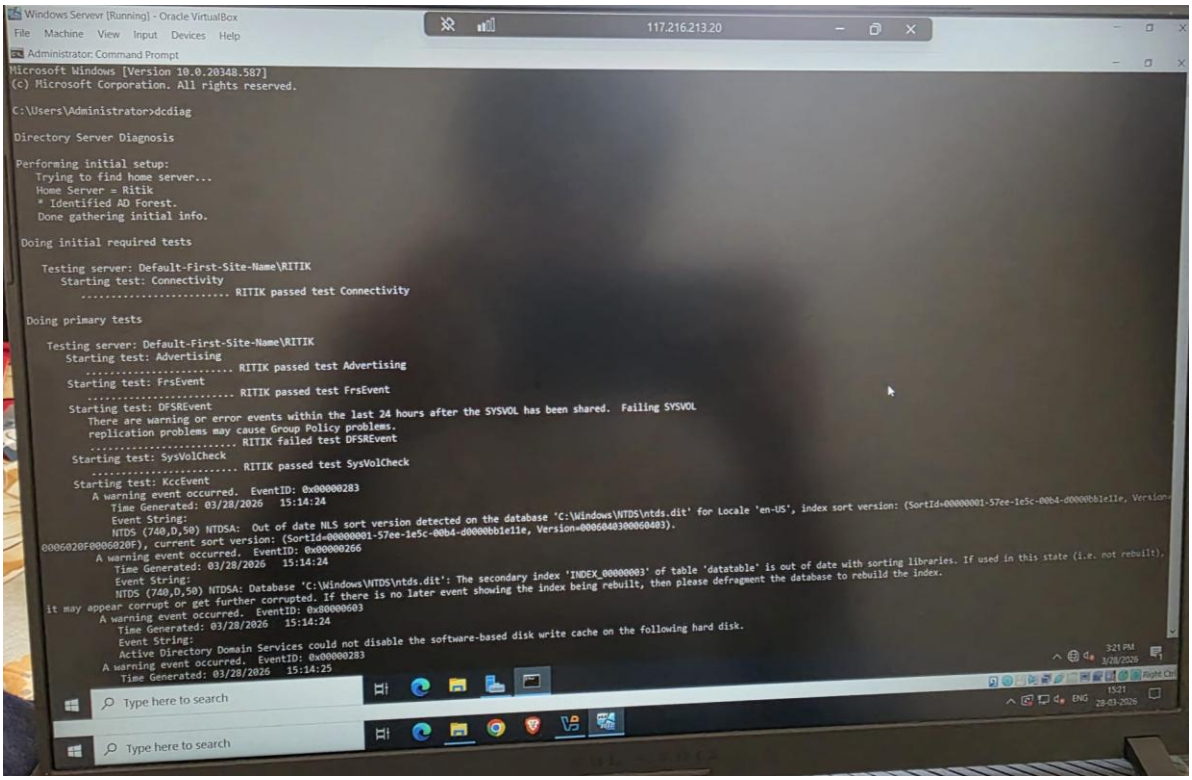
Active Directory Validation

After configuring the Domain Controller, system health and functionality were verified using built-in diagnostic tools.

The dcdiag command was used to analyze the status of the Active Directory services.

This tool performs multiple checks such as:

- Connectivity
- DNS configuration
- Replication status
- SYSVOL availability
- Overall AD health



User Creation in Active Directory

After setting up the domain and verifying its functionality, user accounts were created to simulate real world configuration.

Steps:

1. Opened **Active Directory Users and Computers**
2. Added new users:
    - Administrator (default)
    - meet (custom user)

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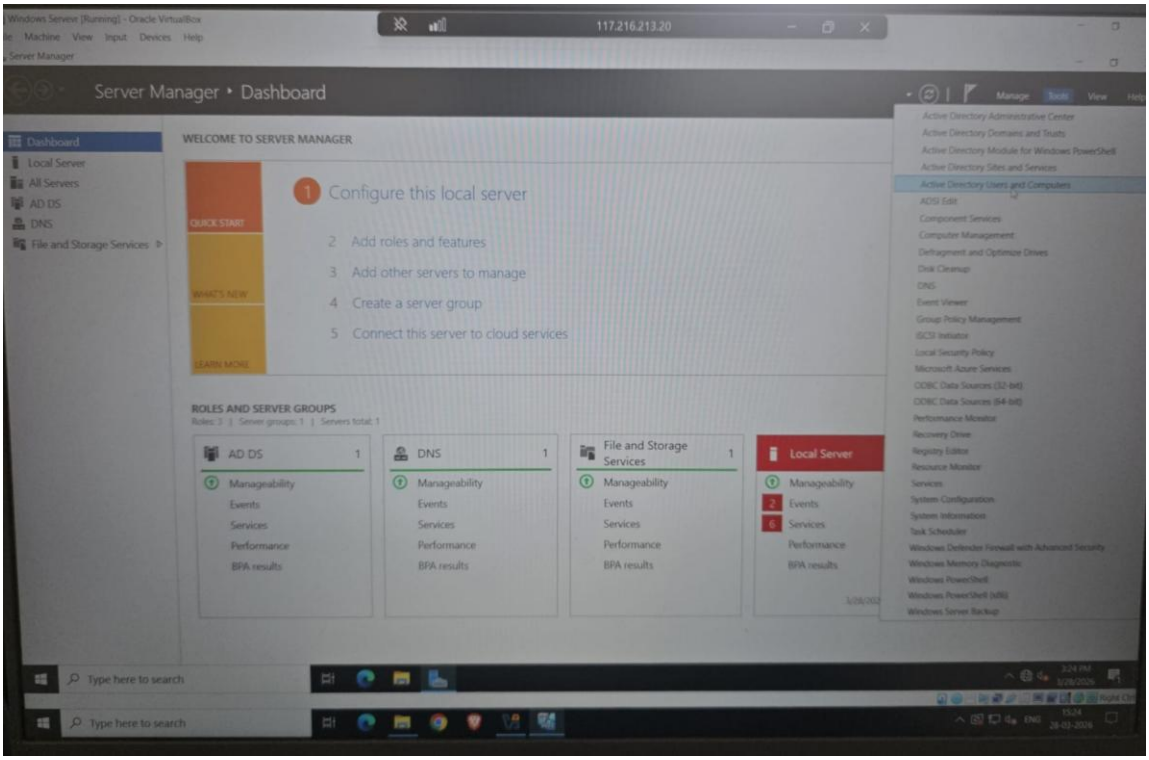

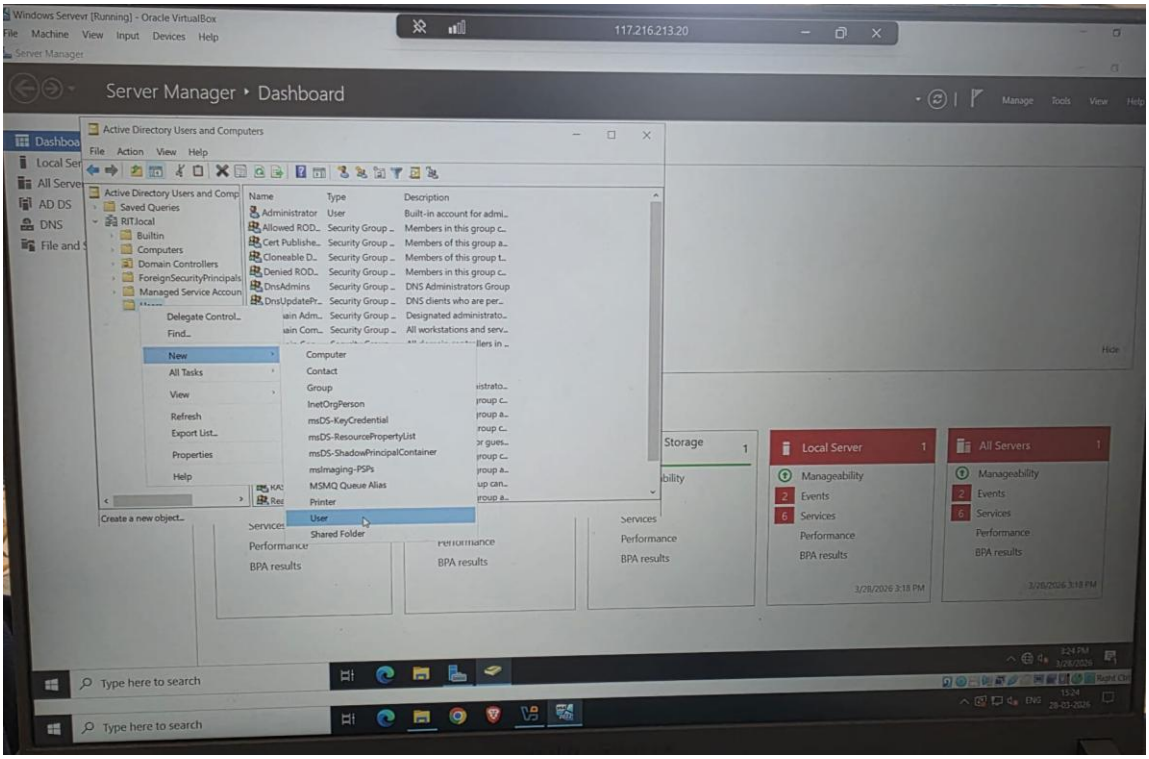

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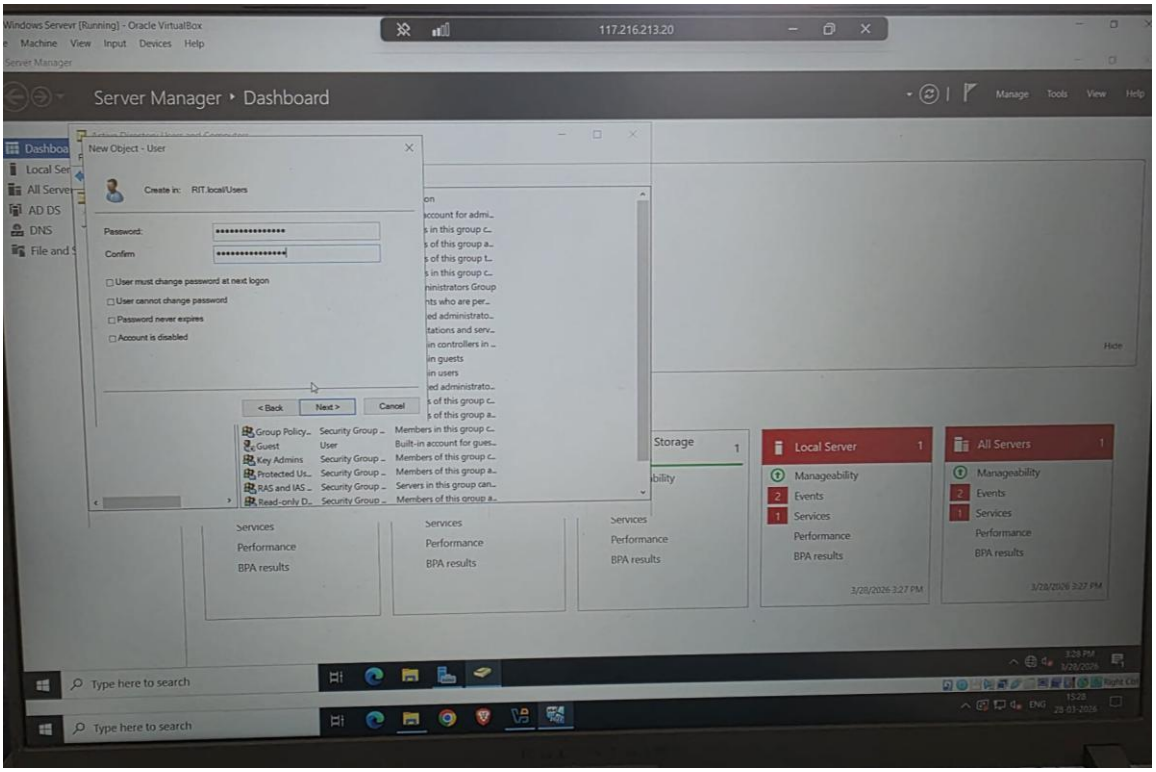

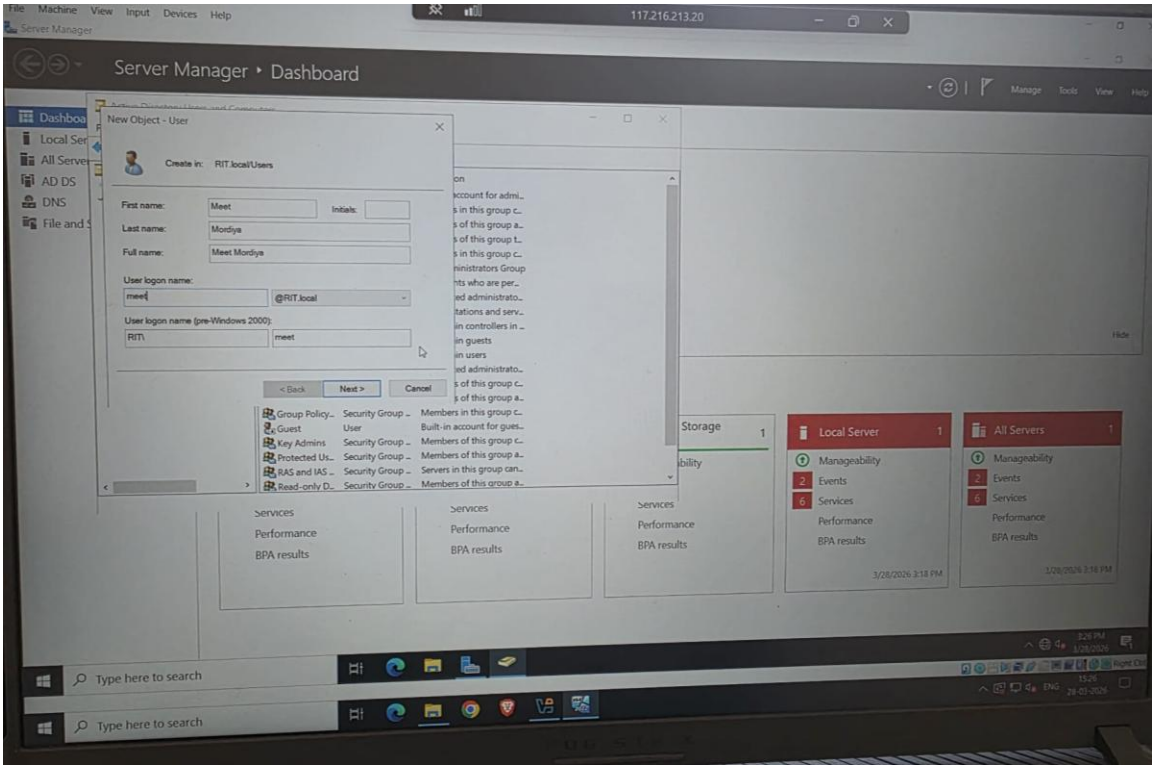

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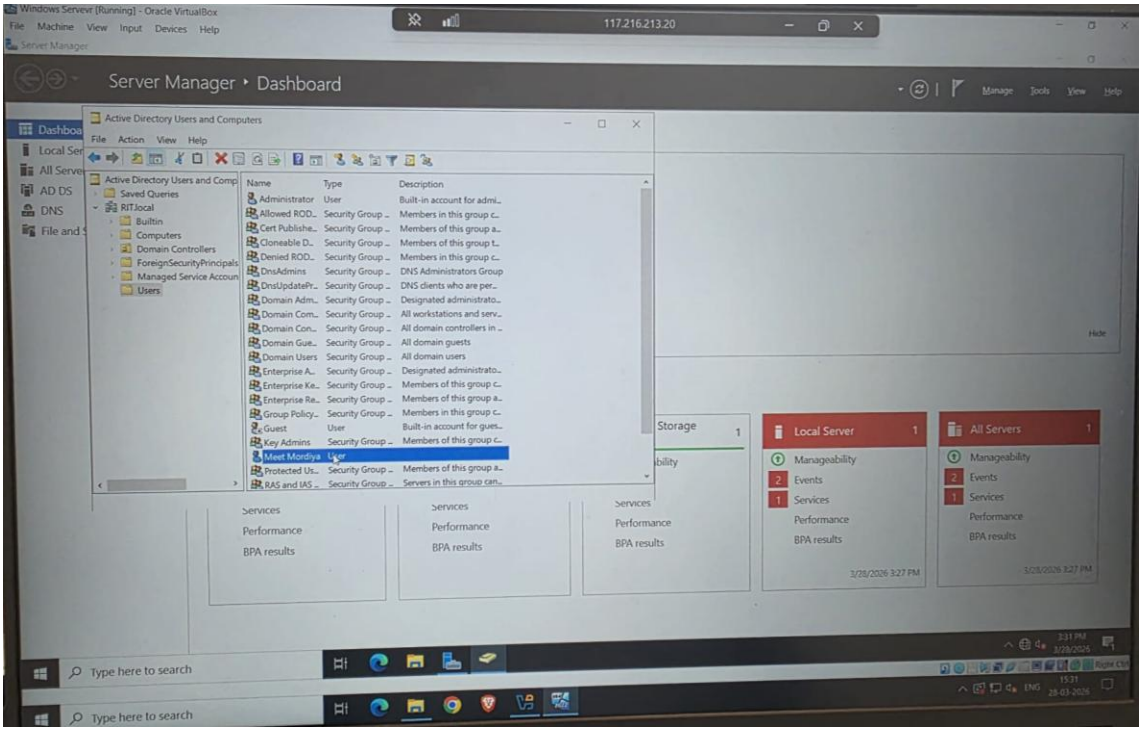

Outcome

At the end of this phase:

- The Domain Controller was fully operational
- Active Directory and DNS were configured
- Users were successfully created
- The environment was ready for client integration

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4. Client Integration (Windows & Ubuntu)

Objective

The objective of this section is to integrate client machines into the Active Directory domain environment and validate centralized authentication across multiple operating systems.

Windows 10 Client Integration

Steps Performed:

1. Installed a windows 10 client on virtual machine
2. Configured DNS on Windows 10 client:
    - Preferred DNS → 172.16.1.3 (Domain Controller)
3. Opened: System Properties → Computer Name → Change
4. Selected: Domain → RIT.local
5. Entered Domain credentials Username and Password
6. Restart the system

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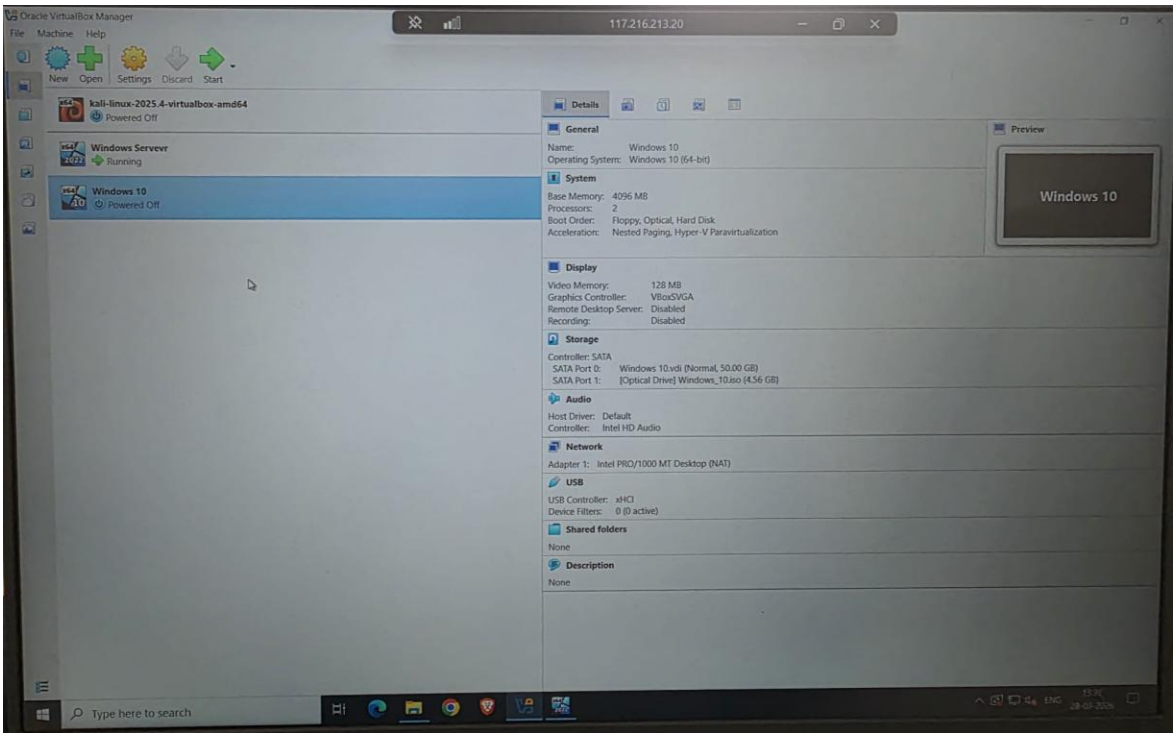

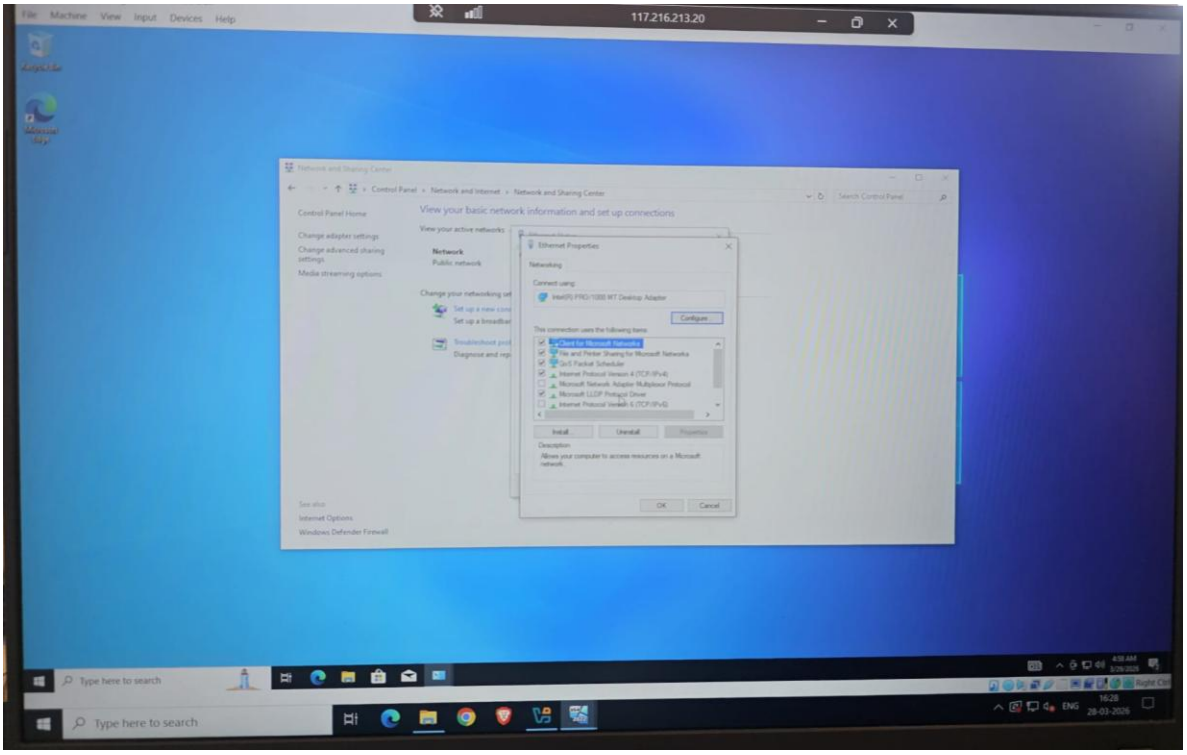

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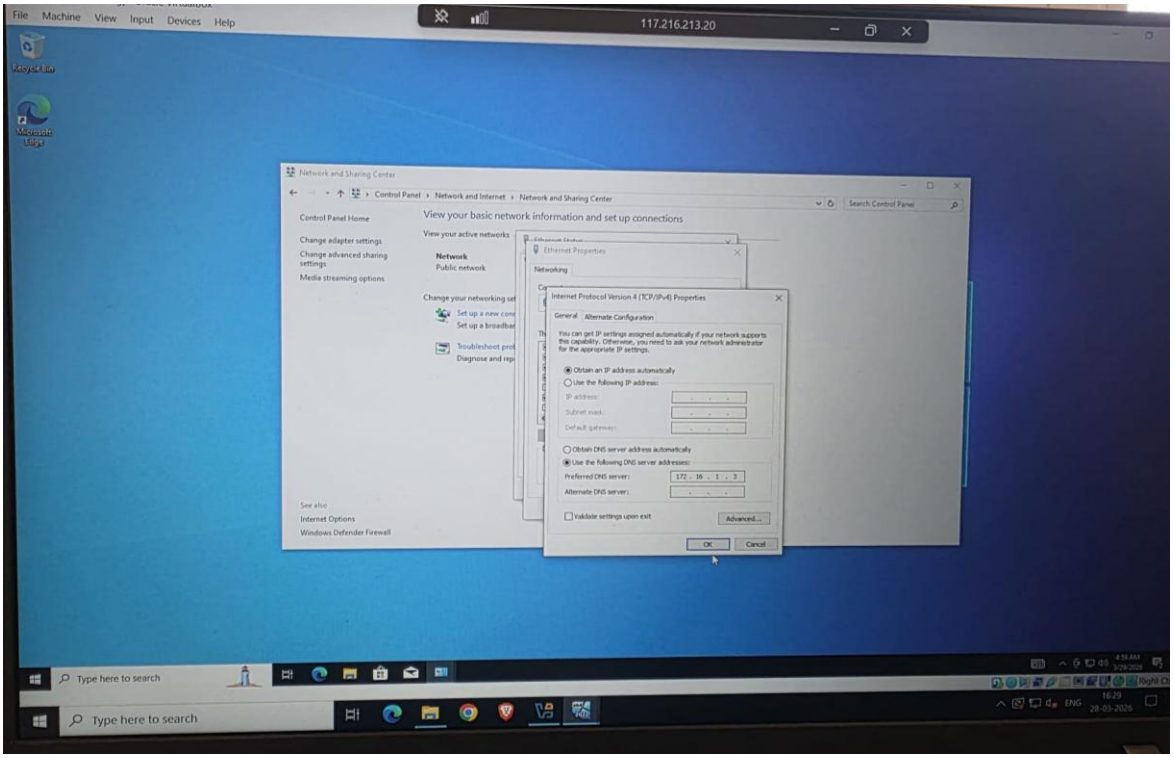

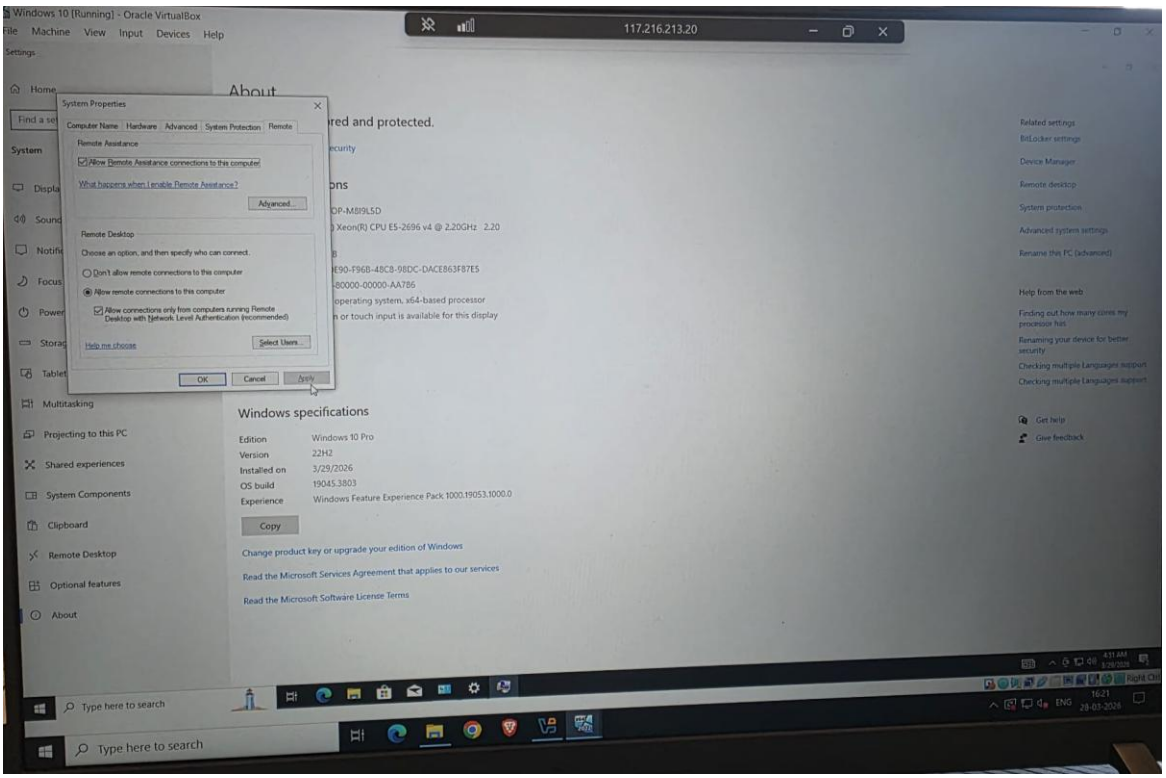

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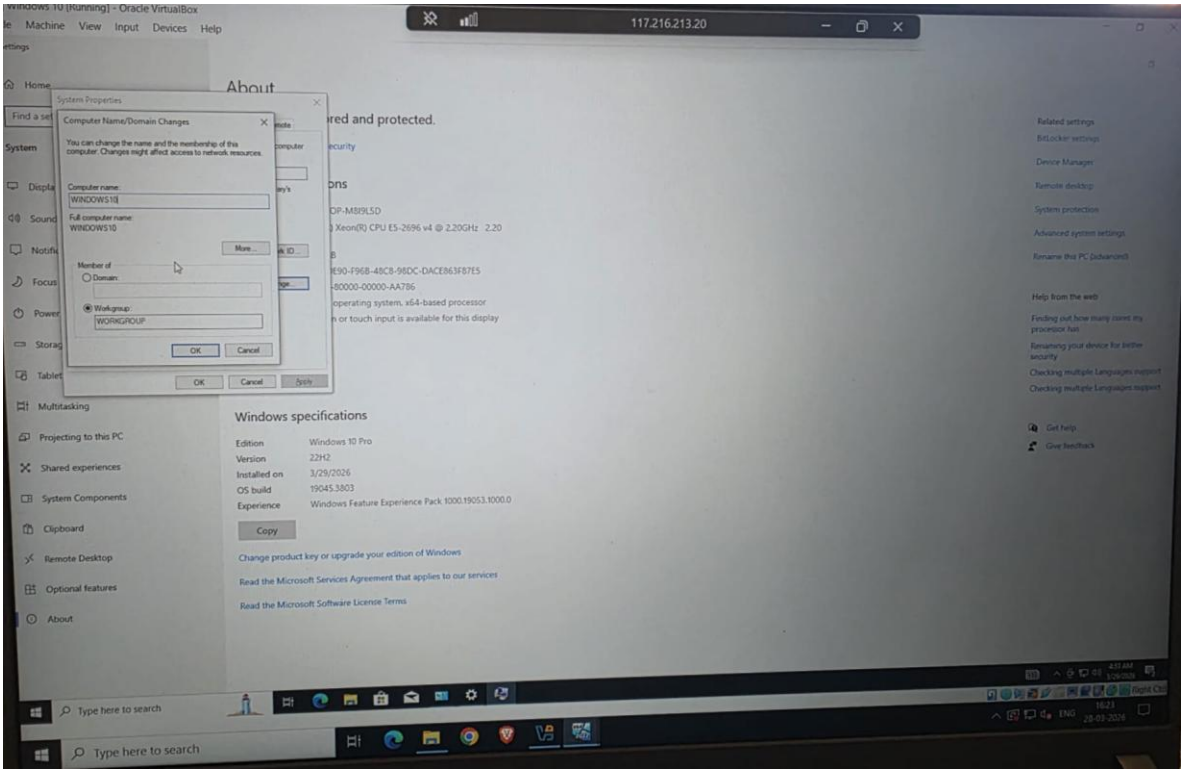

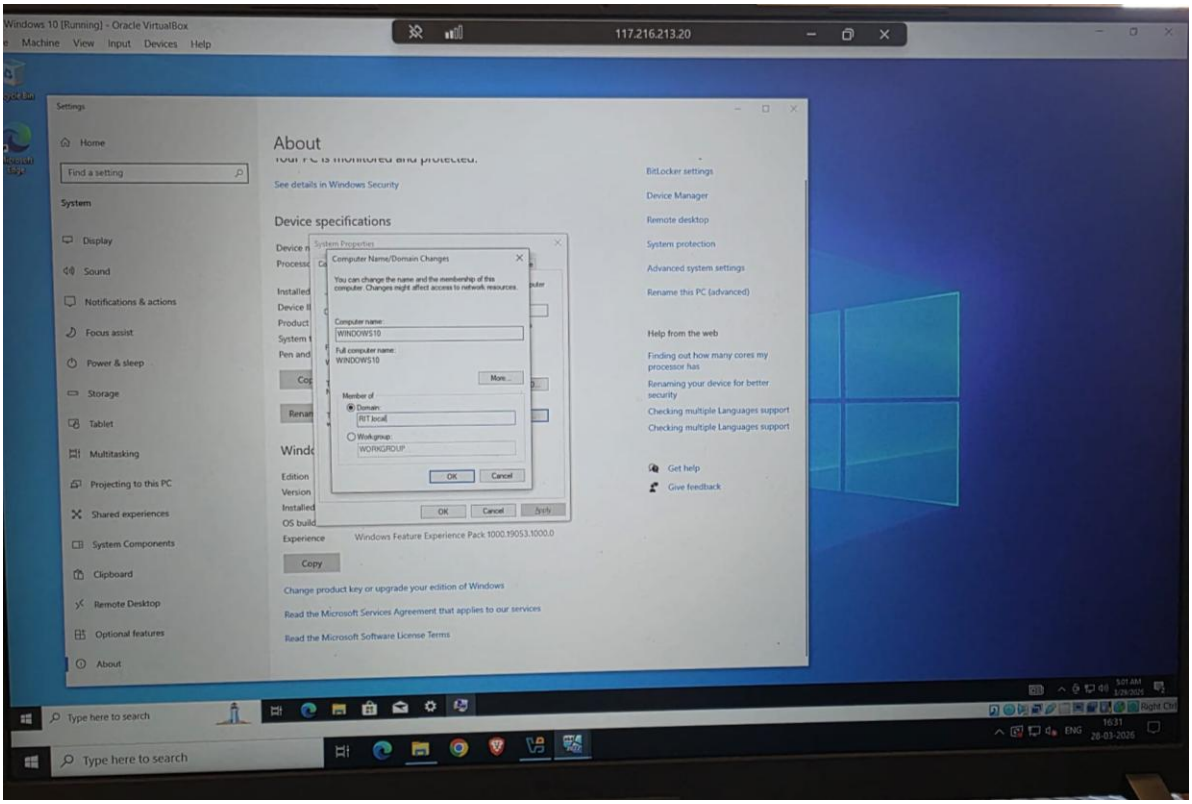

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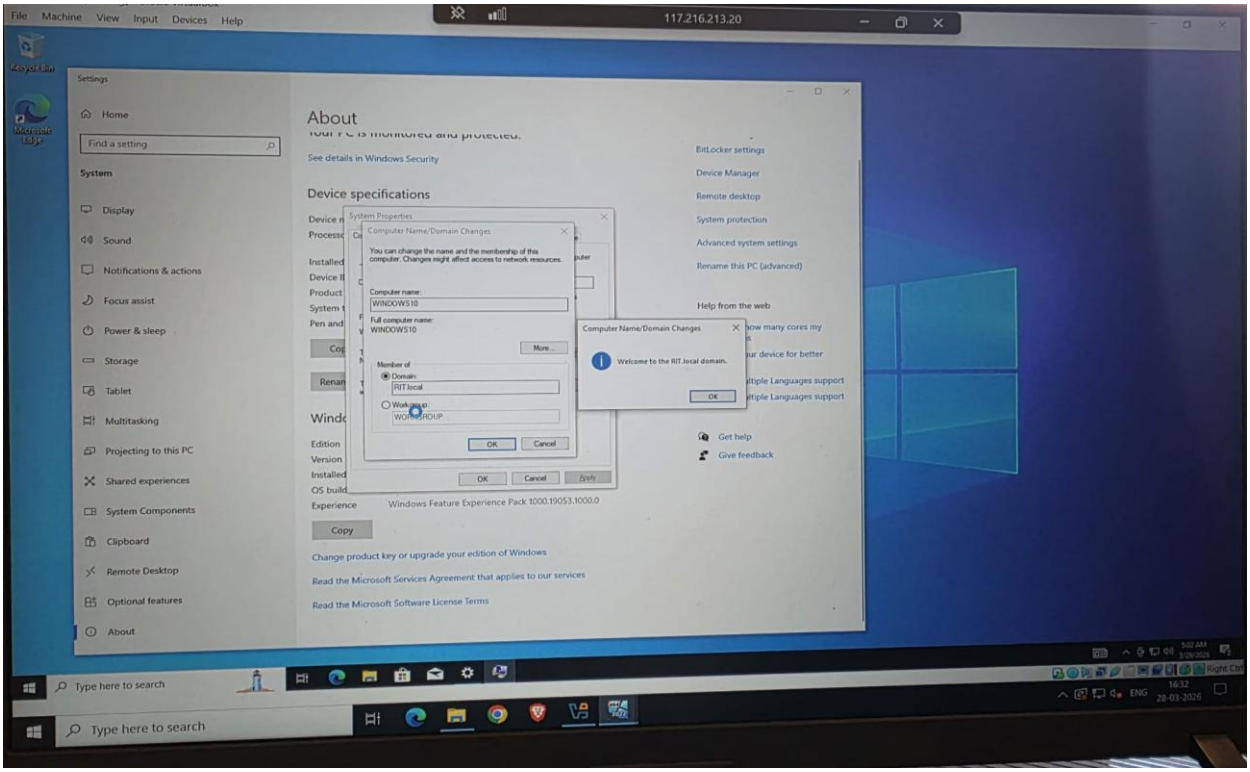

Verification Windows 10 AD setup

Successfully logged in using domain user: RIT\meet

System appeared under: Active Directory → Computers

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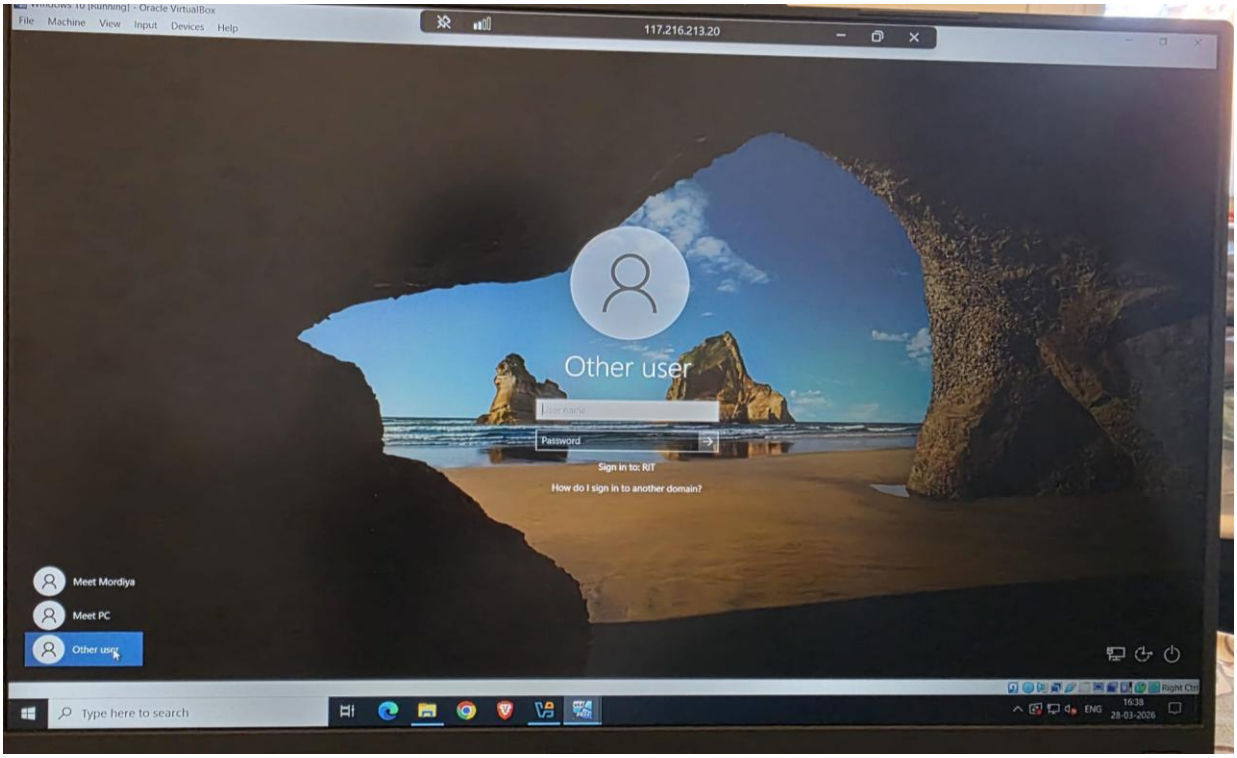

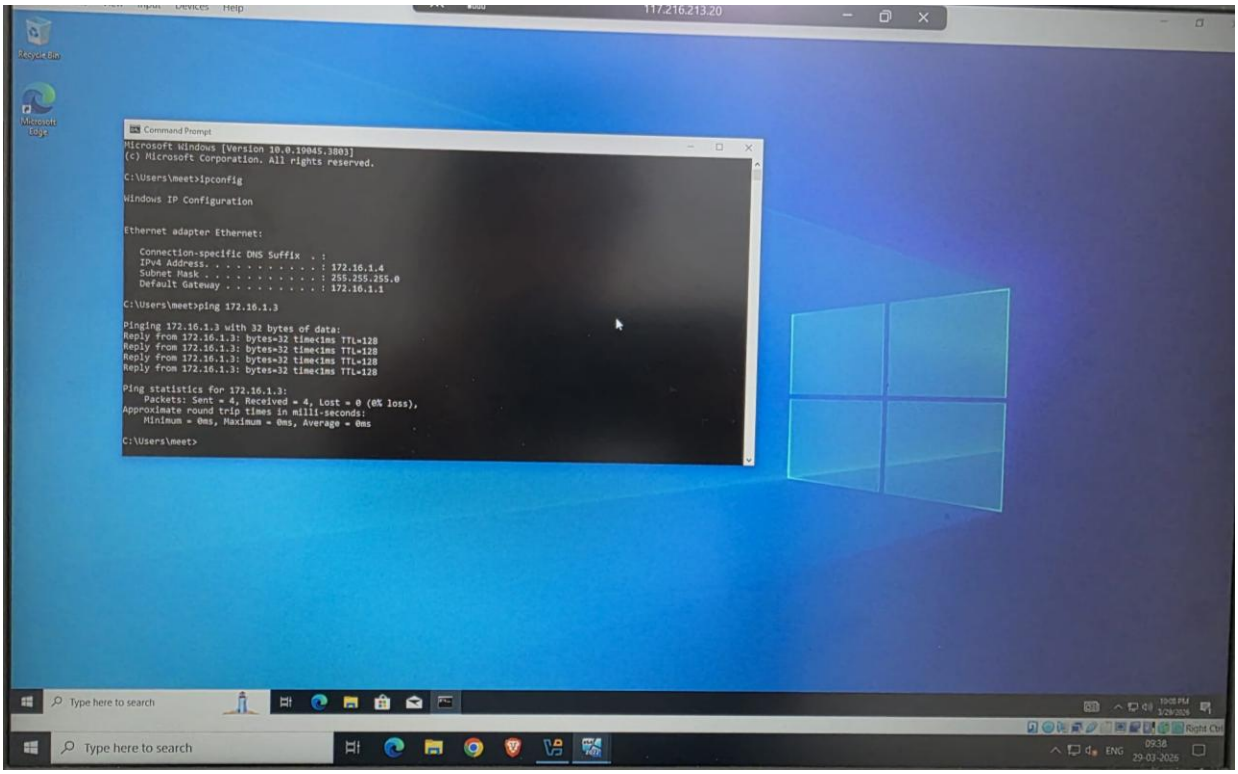

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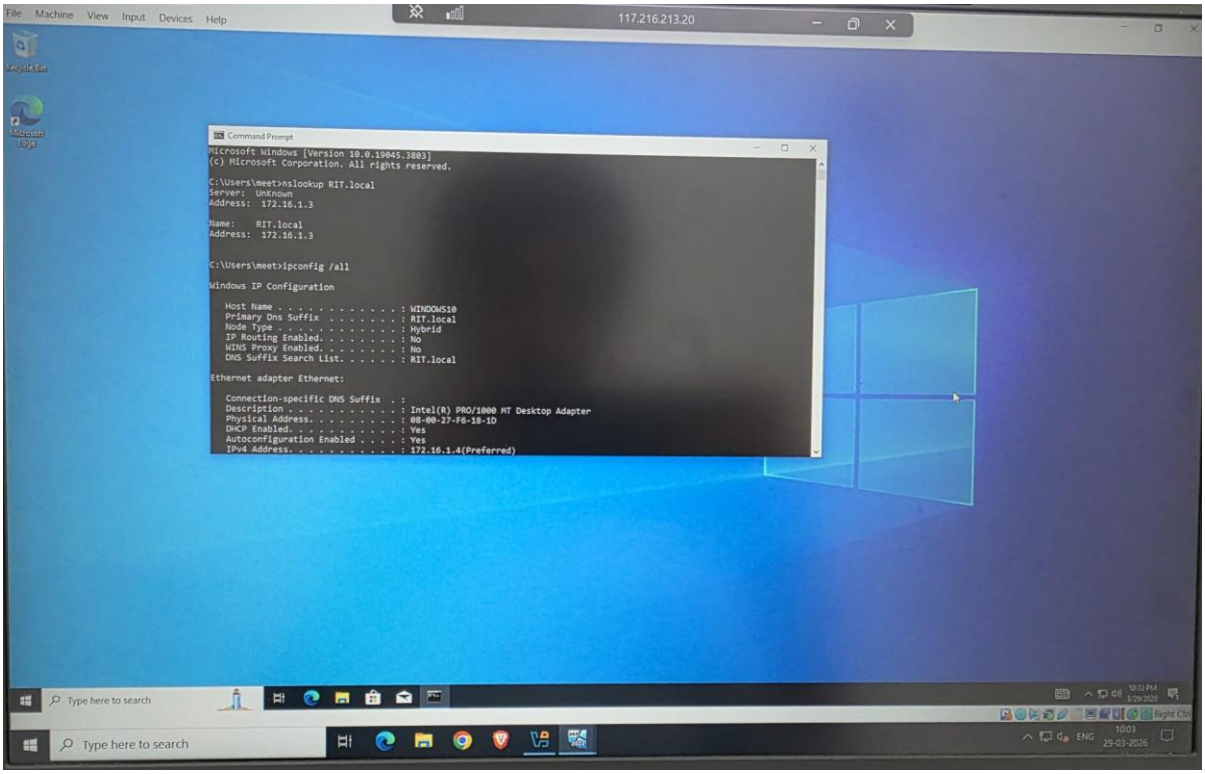

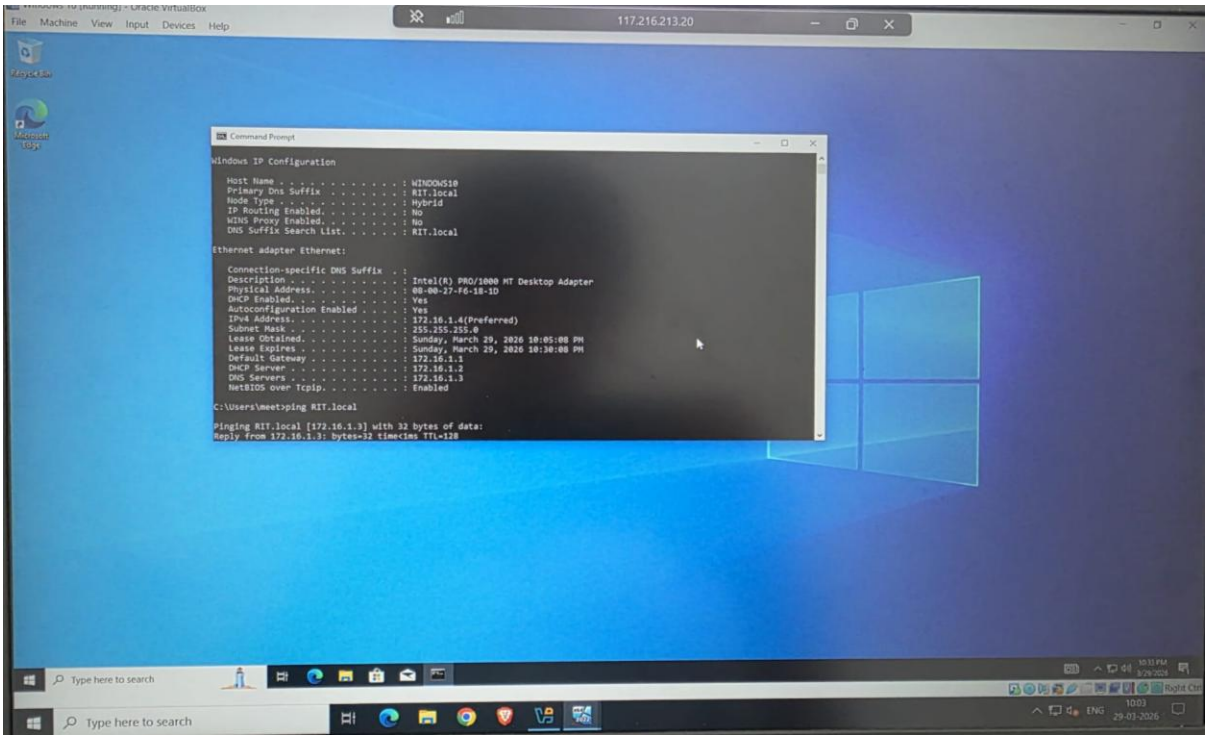

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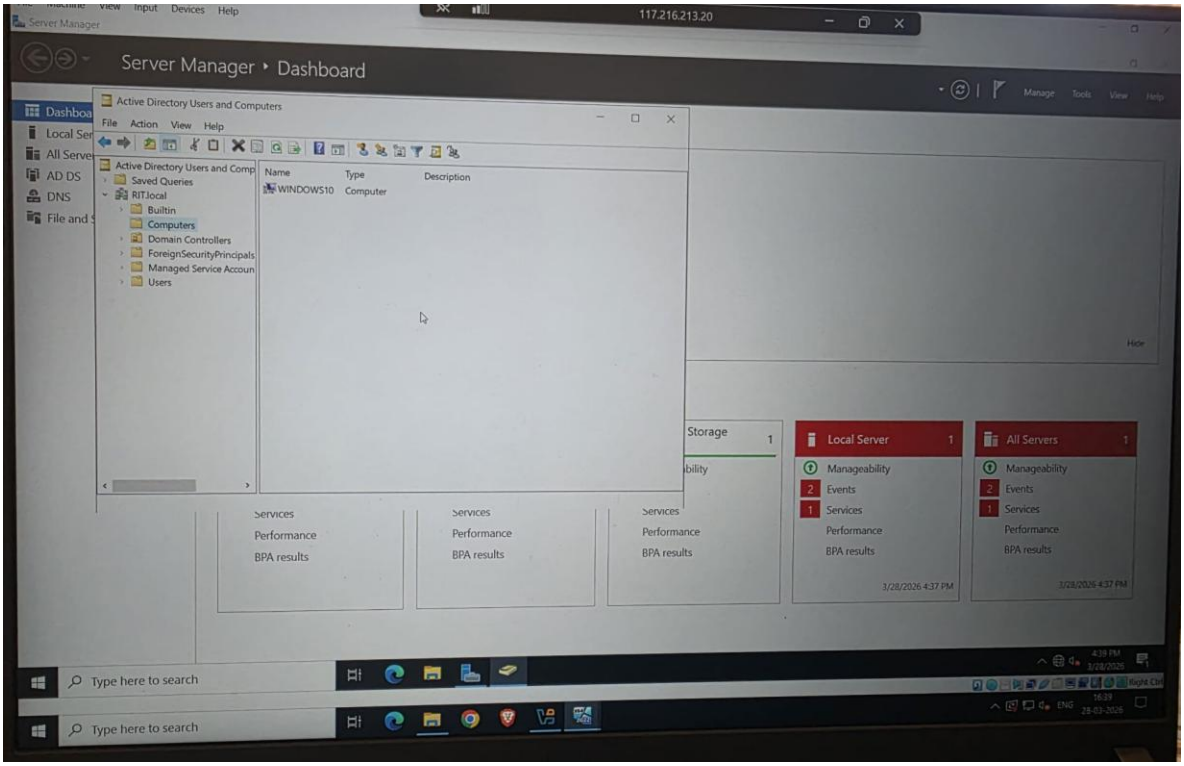

Ubuntu Client Integration

Steps Performed:

1. Installed a Ubuntu Client on Virtual Machine
2. While creating a account select “Use Active Directory”
3. Login to Active Directory

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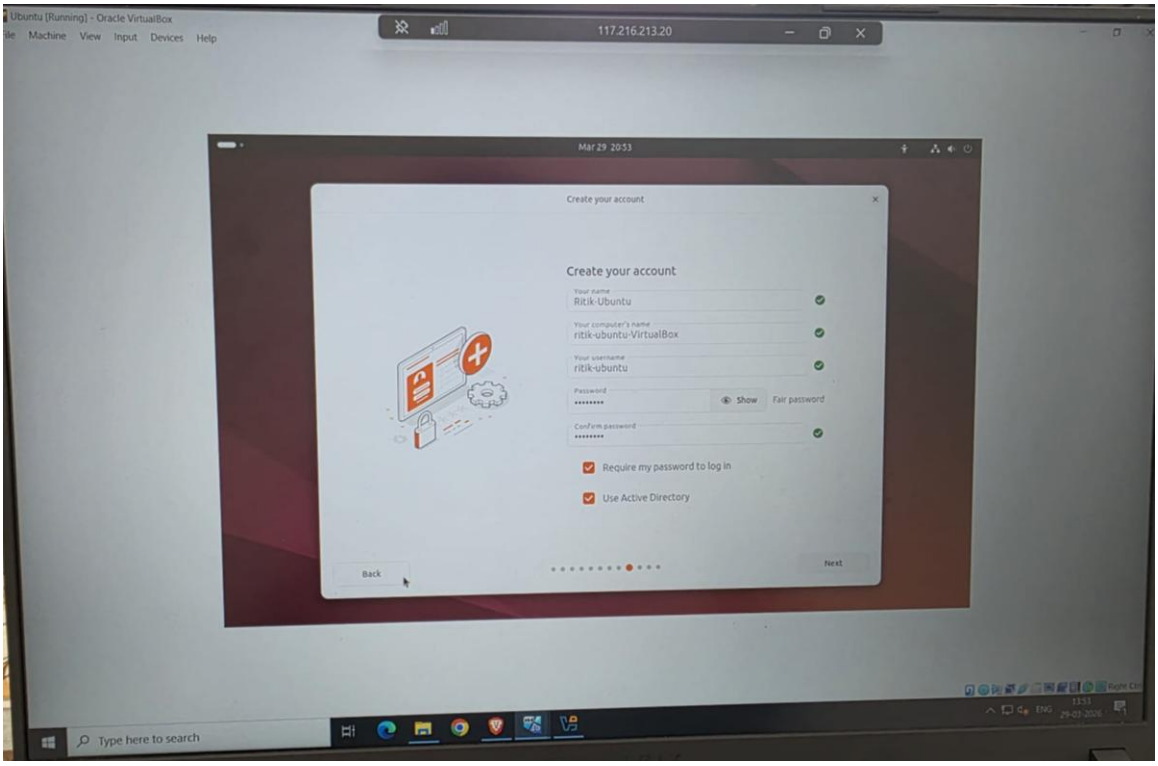

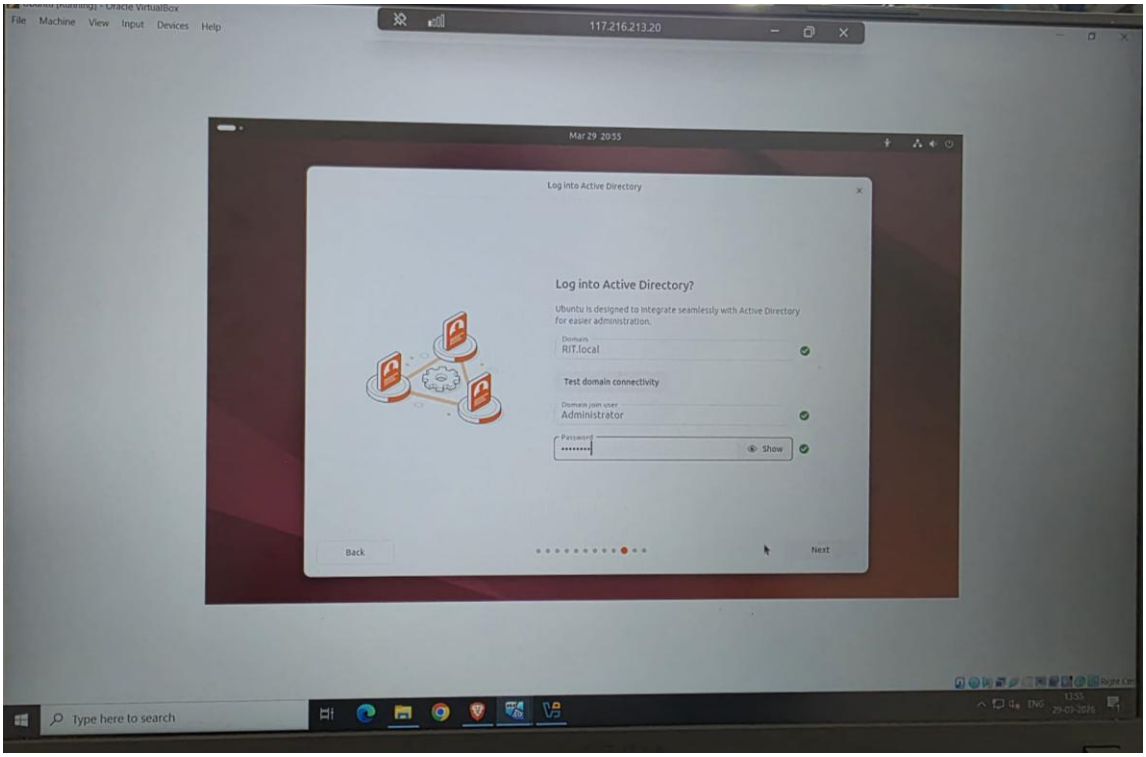

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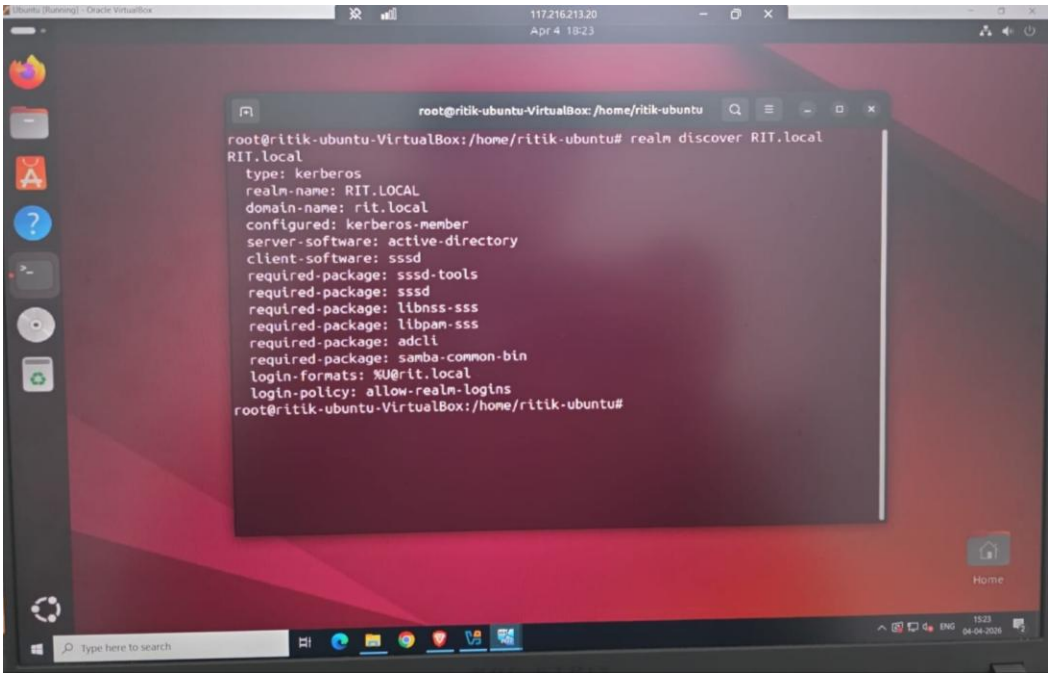
After doing everything correctly if still active directory setup fails then perform the following steps open ubuntu terminal and run the following commands:

1. sudo su
2. apt update
3. apt upgrade
4. apt install gedit
5. apt install realmd sssd sssd-tools libnss-sss libpam-sss adcli samba-common-bin oddjob oddjob-mkhomedir packagekit -y
6. gedit /etc/resolv.conf -> change nameserver IP to Domain Controller or AD server IP
7. realm discover RIT.local



8. realm join -U Administrator RIT.local

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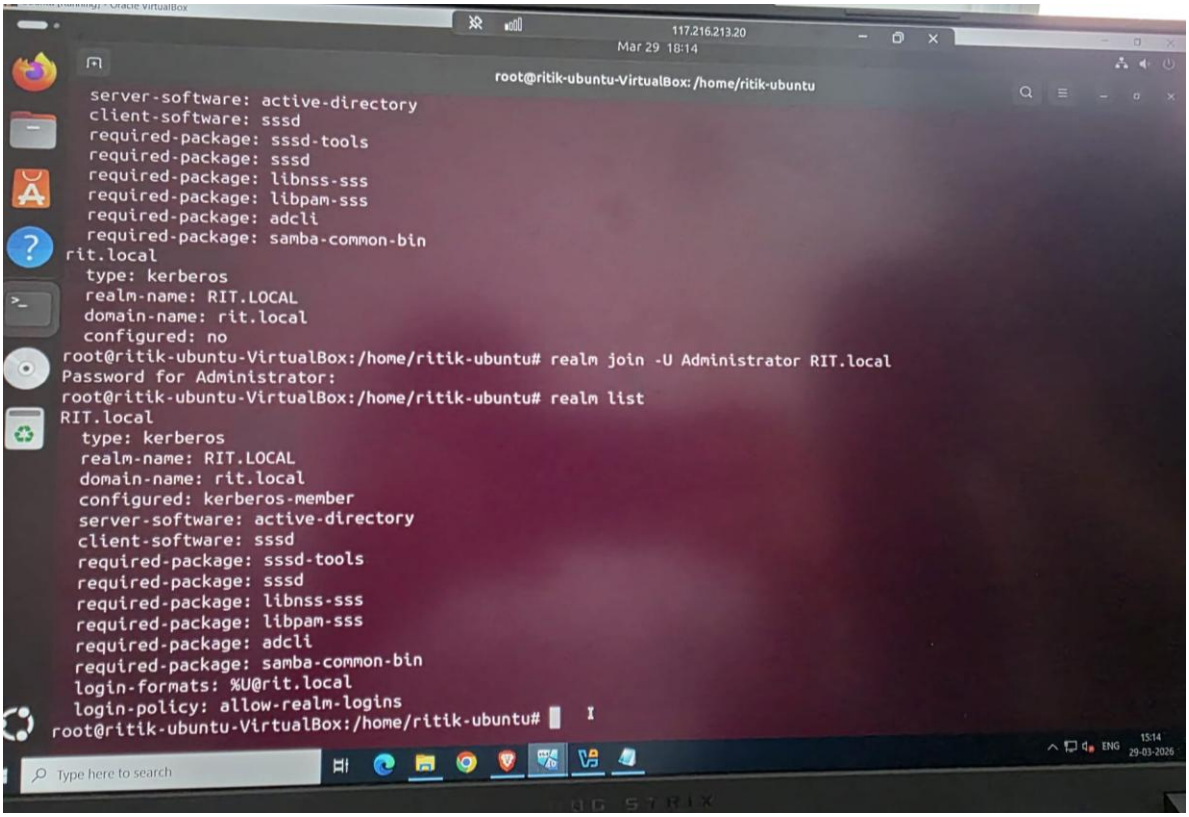

9. realm list
10. sudo pam-auth-update -> press space on: Create home directory on login then press enter

Then restart ubuntu

Verification Ubuntu AD setup

Successfully logged in using domain user: RIT\Administrator

System appeared under: Active Directory → Computers

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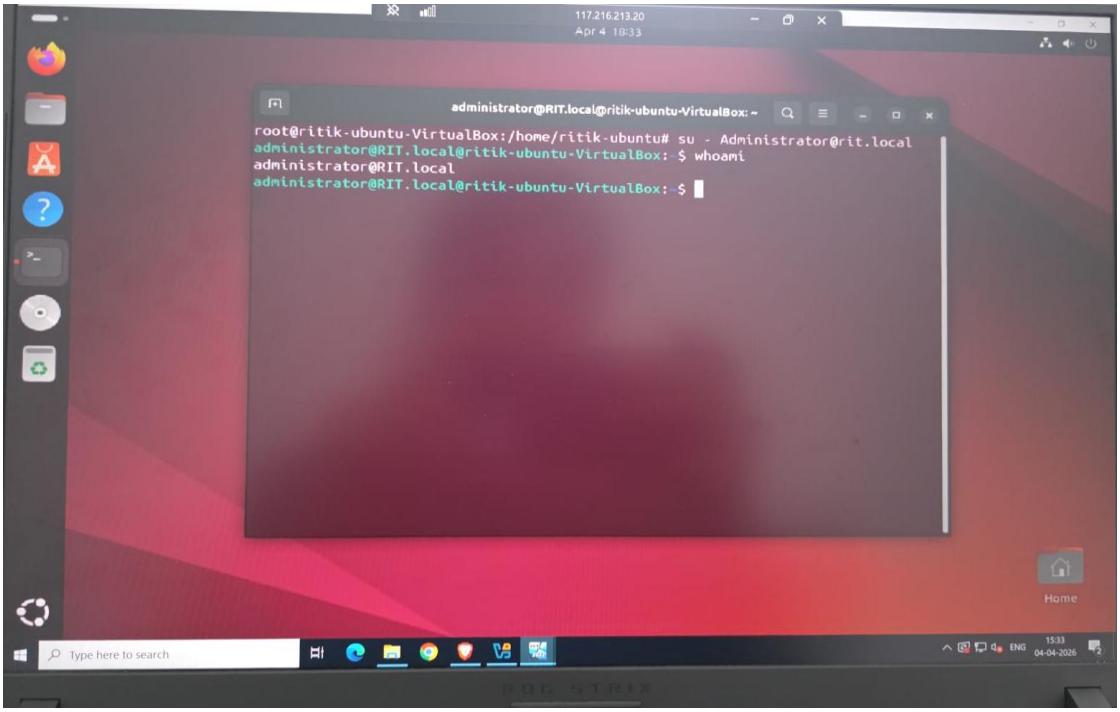

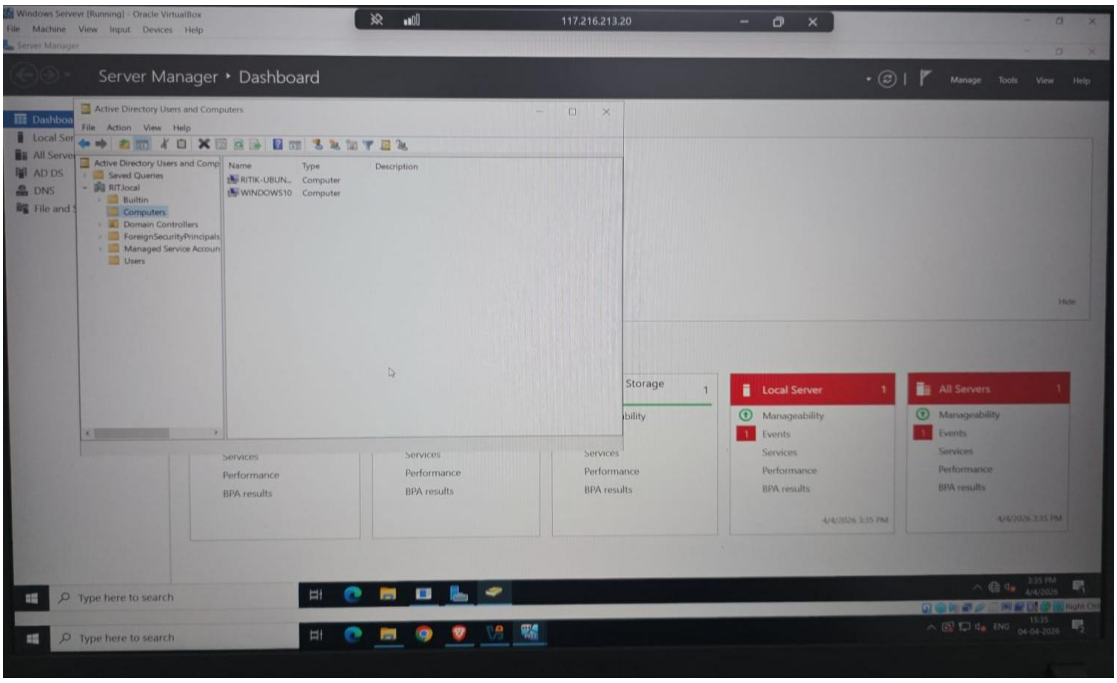

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Key Observations

- Windows integration is seamless and native
- Linux integration requires:
  - ✓ DNS precision
  - ✓ SSSD configuration
  - ✓ Manual troubleshooting

5. Conclusion

Summary

In this project, a complete Active Directory environment was successfully designed and implemented using a virtualized lab setup.

The setup included:

- Windows Server configured as a Domain Controller
- Domain creation: RIT.local
- Integration of multiple clients:
  - Windows 10
  - Ubuntu

Key Achievements

- ✓ Successfully deployed Active Directory Domain Services (AD DS)
- ✓ Configured DNS for domain-based communication
- ✓ Joined Windows and Ubuntu systems to the domain
- ✓ Verified domain authentication across systems
- ✓ Performed system validation using tools like: dcdiag
- ✓ Identified and resolved multiple real-world issues:
  - DNS misconfiguration
  - Linux authentication challenges

Practical Learning Outcomes

This project provided hands-on experience with:

- Enterprise identity management

- Centralized authentication systems
- Cross-platform domain integration
- Network troubleshooting and debugging

It also highlighted the differences between:

- Native Windows AD integration (seamless)
- Linux AD integration (manual and complex)

Real-World Relevance

Active Directory is widely used in enterprise environments for:

- User authentication
- Access control
- Policy enforcement

This project simulates a real-world corporate infrastructure where multiple systems rely on a centralized identity provider.

Future Work

- Implement Group Policy Objects (GPO)
- Integrate monitoring tools (Wazuh) for security analysis
- Simulate real-world attack scenarios (e.g., brute force detection)